ZS-A19

Z-Spin Boundary Tension as Geometric Dust: A Rank-Absorption No-Go and a Boundary-Charge (ZHCS) Closure Lifting the 32/121 Dark-Matter Ratio to DERIVED-CONDITIONAL

Kenny Kang
Z-Spin Cosmology Collaboration
June 2026 — Astrophysics theme [ZS-A] · Paper 19 · ZS-A19 v2.0
GitHub: *https://github.com/KennyKang-git/zspin*

Verification: 60/60 checks PASS (representation theory, arithmetic, a CAMB acoustic-peak computation, four review-driven obstruction proofs, four ZHCS structural checks, and the ZHCS-1 polyhedral closure computation). v2.0 carries out the deep-exploration closure of the two gates v1.5 left open. **New in v2.0 — ZHCS-1 is now DERIVED**: the 38-node seam graph is connected (rank $L_\Gamma = 37$) — verified by building the actual truncated-icosahedron face graph (32 nodes, connected) and the cube face graph (6 nodes, connected) and computing the Laplacian rank; the cross-coupling is the PROVEN rank-1 β_0 Z-mediation ($L_{XY} \equiv 0$, $\kappa^2 = A/Q \neq 0$), and the Euler null mode of the TI vertex-face incidence (ZS-F2 Lemma 4.5, PROVEN: $v_p = -2v_h$, $(\Sigma v)^2 = 4/17$) is nonzero on all 32 faces, giving the cross-block full support. **ZHCS-2 is now DERIVED-CONDITIONAL**: the ZS-F0 BV-BFV modified quantum master equation ($\Omega_{\partial\Sigma}\, \psi = 0$) plus the BRST Hodge decomposition (harmonic = closed ∩ co-closed) plus the seam J as a boundary constraint operator ($\|[J,L]\| \approx 2.94 \neq 0$, PROVEN) supply the graph-harmonicity scaffold; the single residual is the identification $\Omega_{\partial\Sigma} \cong d_\Gamma$ on the 38-face basis. **Consequently the 32:6 ratio rises from "PROVEN-CONDITIONAL on two OPEN gates" (v1.5) to DERIVED-CONDITIONAL** — conditional now on the single ZHCS-2 identification, not on two open problems; A19.NG2 is respected throughout (the common rescaling $p_\star \to \alpha\, p_\star$ leaves the ratio invariant). Retained — A19.NG2 (PROVEN NO-GO, bulk rank-lock absorbed); the dust EOS is IMPORTED-PROVEN; the CAMB result is OBSERVATIONALLY CONSISTENT, not VERIFIED; the v1.3 "single node" is RETRACTED (A19.NG1). ZS-A18 NO-GO frozen. Zero new fitted Z-Spin parameters. AN-A19.1, AN-A19.2 EXECUTED, PASS.

§0. Abstract

ZS-A18 (v1.5, frozen) established that a massless Z-Spin Goldstone gradient cannot serve as the recombination cold dark matter (CDM) responsible for the third CMB acoustic peak: a smooth massless field has equation of state $w \in \{-1/3, +1\}$, never $w = 0$. This paper does not contest that NO-GO. Instead it proposes a different carrier. We abandon the premise — shared by every previously blocked route — that dark matter is a collection of objects (Planck-mass vortex cores, current-carrying vortons, primordial-black-hole relics). That premise is responsible for the over-closure, intermediate-formation-scale (~10^9 GeV), and current-carrier obstructions, all of which we show are artifacts of the particle ontology rather than of the physics.

We instead implement ZS-F2's original definition of CDM as a geometric boundary tension, not a particle species, through a conserved pullback dust. Three external PROVEN mechanisms — the relativistic pullback perfect-fluid action, the mimetic constraint, and the projectable integration-constant of Hořava-type gravity — yield exactly $w = 0$, $c_s^2 = 0$, $\rho \propto a^{-3}$ with no microscopic particle relic abundance and no formation temperature required, while preserving $m_\theta = 0$ (the ZS-S3 Goldstone theorem and the ZS-A1 ε-Halo). The over-closure obstruction is thereby not evaded but removed: its calculation (Kibble density × Planck mass) has no object to which it applies. The dust does carry a

conserved comoving number/entropy current and an energy-per-element normalization μ; that μ is an independent equation-of-state input and is not yet derived from ZS-F2, so this paper introduces no new fitted parameter but leaves μ, the state selection, and the action-level origin OPEN.

We then give the **equivariant boundary-rank–trace theorem**: *F(TI) = 32* $\Longrightarrow$ *rank P_BT = 32* $\Longrightarrow$ *τ_Q(P_BT) = 32/121*, via the A_5 deleted-icosahedral representation $V_{11} = 3 \oplus 3' \oplus 5$ and the truncated-icosahedron face module F_TI $\rightarrow$ $\mathrm{End}(V_{11}) \cong M_{11}(\mathbb{C})$. We prove the "32 charge" is neither a spatial topological charge (NO-GO: ∂TI $\cong S^2$, so $H^2(S^2,\mathbb{Z})$ has a single generator and face subdivision changes the count) nor a metric tension (NO-GO: the 32 faces are non-congruent — 12 pentagons and 20 hexagons — so area weighting gives $\neq$ 32/121), but an equivariant projection-class charge. No equipartition assumption enters the algebraic identity (it is the definition of the canonical normalized trace). Two limitations are explicit: the rank-32 projector is not unique — every rank-32 projection has trace 32/121, so the physics lies in why the dynamics selects the TI face-module image — so P_BT is a candidate, not yet a canonical, projection; and physical use of the identity (Ω_cdm = 32/121, not merely τ_Q = 32/121) requires the tracial-state and energy-degeneracy conditions of BT-C below.

The central physical bridge — that the gravitational equilibrium state restricts to the canonical trace (**Condition BT-C**) — is **OPEN**. It is of the same kind as ZS-F23 Condition C (a physical state realizing a canonical trace), but it is *not* the same trace on the same algebra: **Theorem A19.NG1** (PROVEN NO-GO) shows that the unique faithful unital embedding A_ZS = $M_3(\mathbb{C}) \oplus \mathbb{C} \oplus M_5(\mathbb{C}) \rightarrow M_{11}$ forces center weights (3,3,5)/11, never the ZS-F23 equilibrium weights (3,2,6)/11, so a single normalized matrix trace cannot induce both. The earlier claim that one theorem closes both is therefore withdrawn; we replace it with a finite set of action-level and operator-algebraic gates, led by BT-C but not reducible to it, together with the registered tensions dim(**Z**) = 2 $\notin V_{11}$ (the A_5 action does not respect the (2,3,6) sector blocks) and the A_ZS/M_{11} trace incompatibility (NG1). This is a derivation ***program***, not a closure: it converts the third-peak CDM problem from one of numbers and energy scales into a finite, sharply stated set of operator-algebraic and action-level theorems.

In v1.2 we present a more corpus-native action-level route that bypasses the hardest of these gates: **Boundary-Rank-Locked Z-Clock Dust**. A constrained Z-clock field T_Z with a non-propagating multiplier λ_c gives, at the action level, exact pressureless dust ($w = c_s^2 = 0$, $\rho \propto a^{-3}$) — this equation of state is IMPORTED-PROVEN ordinary clock dust. The ansatz further proposes a rank-32 seam projector P_c (built from the X-democratic and Z-odd directions, free of the A_5/(2,3,6) tension) and a baryon-anchored boundary-matching term that, combined with the corpus baryon asymmetry η_B = (6/11)^35 and the standard BBN relation, writes the h-free numbers ω_c = (32/6) ω_b = 0.119112 (−0.89σ from Planck) and ω_c/ω_b = 16/3 (−0.48σ), replacing ZS-F2's h-circular ω_c = (32/121)h^2. Whether these numbers are derived or merely imposed is exactly what v1.4 re-examines, immediately below. The §5 rank–trace and Theorem A19.NG1 are retained as an independent geometric cross-check and a standing no-go; the ZS-A18 NO-GO is unmodified.

v1.3 tested these gates by computation (a CAMB acoustic-peak run, the S_3 uniqueness of the seam mode, the δ_c = δ_b implication, and the i-tetration clock norm). v1.4 incorporates an external technical review that identified a decisive obstruction overlooked by that testing, and records the consequences honestly. **The obstruction (Theorem A19.NG2, PROVEN):** for a scalar clock T_Z and scalar multiplier λ_c, the clock action depends on the seam projector only through its trace, Tr P_c = 32, which is removed by the redefinition $\lambda_c \to \lambda_c/32$; the resulting theory is ordinary single-clock mimetic dust, and any rank gives the same theory. The rank-32 lock therefore has no effect in the bulk action — it can be physical only as a quantized boundary Noether charge or via operator-valued dynamics, both of which remain OPEN. **Consequences.** The geometric dust equation of state ($w = 0$, $\rho \propto a^{-3}$) is IMPORTED-PROVEN ordinary clock dust; the ratio 32:6 is imposed by the boundary lock S_match (which can impose any ratio), not derived; equipartition (Theorem A19.5) is an IMPORTED-PROVEN thermodynamic precedent whose Z-Spin application is OPEN, since a rank-to-energy identity needs mode-energy degeneracy, a relativistic-to-dust freeze-out preserving 32:6, and the baryon thermal history; the dust clock as the Z-anchor proper time is HYPOTHESIS-strong/OPEN, since a worldline proper time is a global scalar clock only under hypersurface-orthogonality and a smooth global foliation; and the CAMB result is OBSERVATIONALLY CONSISTENT (a consistency check assuming standard cold-dark-matter transfer, TT third peak only), not VERIFIED. **Retraction.** The v1.3 claim that three routes converge on a single emergence node is withdrawn: the paper's own Theorem A19.NG1 proves the channel trace (Condition BT-C) and the ZS-F23 modular weights (Condition C) are distinct traces, and the honest residual is a set of 6–8 distinct open gates (§9.6). The net effect is a sharp contraction: the dust EOS is rigorous, a new self-no-go bounds the mechanism, and the abundance and origin remain an explicit action ansatz gated by several distinct open problems. The ZS-A18 NO-GO is intact and no new parameter is fitted. Adding A19.NG2 raises the honesty of the program; recording it is the result.

v1.5 adds the most conservative route by which those gates might close (§10), a boundary-charge construction we call **Z-Seam Harmonic mass-Charge Splitting (ZHCS)**. Rather than placing the rank in the bulk action (which A19.NG2 forbids), ZHCS lets a single conserved boundary canonical charge p live on a connected 38-node graph of the 32 truncated-icosahedron cold faces and the 6 truncated-octahedron baryon faces; if the ZS-F0 boundary variation forces p to be graph-harmonic ($d_\Gamma p = 0$) and the graph is connected, then p is constant and the two matter charges are $Q_c = 32 p_\star$ and $Q_b = 6 p_\star$, so $Q_c/Q_b = 32/6$ as the **rescaling-invariant ratio of two projections of one parent charge** — A19.NG2 removes the absolute scale but not this ratio, and 32:6 is never written into the action. The bulk uses ordinary Brown–Kuchař dust, which eases the clock problem (the Z-anchor becomes a Jacobian locator, not the whole congruence) and supplies a gauge-invariant $S_{cb} = 0$. This reduces G1–G8 to four minimal theorems: ZHCS-1 (the graph is connected, rank $L_\Gamma = 37$ — an immediately computable finite-matrix fact), ZHCS-2 (the ZS-F0 variation yields $d_\Gamma p = 0$ — the real action-level bottleneck), and ZHCS-3/4 (canonical gluing to the dust momentum, and a single adiabatic parent clock). The rank-ratio theorem is PROVEN conditional on ZHCS-1 $\wedge$ ZHCS-2; the program as a whole is HYPOTHESIS-strong and introduces no new fitted number. If ZHCS-1 and ZHCS-2 close, the third-peak mechanism could be raised, for the first time, above DERIVED-CONDITIONAL.

v2.0 carries out that closure by deep corpus search and explicit computation. **ZHCS-1 is now DERIVED**: building the truncated-icosahedron face graph from the icosahedron (32 nodes — 12 pentagons of degree 5 and 20 hexagons of degree 6 — connected, Laplacian rank 31) and the cube face graph (6 nodes, connected, rank 5), the 38-node seam graph has Laplacian rank $L_\Gamma = 37$; the cross-coupling is the PROVEN rank-1 β_0 Z-mediation ($L_{XY} \equiv 0$, $\kappa^2 = A/Q \neq 0$), and the Euler null mode of the TI vertex-face incidence (ZS-F2 Lemma 4.5, PROVEN: $v_p = -2v_h$ on all 12 pentagons, v_h on all 20 hexagons, $(\Sigma v)^2 = 4/17$) is nonzero on every face, so the cross-block has full support and even a single cross-edge suffices for connectivity. **ZHCS-2 is now DERIVED-CONDITIONAL**: the ZS-F0 BV-BFV modified quantum master equation $(\hbar^2\Delta_\mathcal{V} + \Omega_{\partial\Sigma})\psi = 0$ (IMPORTED-PROVEN, Cattaneo–Moshayedi–Wernli), the BRST

Hodge decomposition giving harmonic = closed ∩ co-closed (IMPORTED-PROVEN, Malik), and the seam J as a boundary constraint operator ($\|[J,L]\| \approx 2.94 \neq 0$, PROVEN) together supply the graph-harmonicity scaffold $p \in \ker L\_\Gamma$; the single residual is the explicit identification of the boundary BFV operator $\Omega\_\partial\Sigma$ with the graph coboundary d_Γ on the 38-face basis. **Therefore the 32:6 ratio rises from PROVEN-CONDITIONAL on two OPEN gates to DERIVED-CONDITIONAL**, conditional now on the single ZHCS-2 identification rather than two open problems. This is a genuine advance, not a closure: ZHCS-2's operator identification and the perturbation theorems ZHCS-3/4 remain, so the result is reported as DERIVED-CONDITIONAL and not as an unconditional derivation.

§0.1 Epistemic Status Legend

Table 0. Epistemic status legend used in this paper.

Status	Definition
LOCKED	Core constant derived and fixed upstream; no downstream paper may modify. Here: A = 35/437, Q = 11, (Z, X, Y) = (2, 3, 6).
PROVEN	Mathematical theorem; follows from standard mathematics or corpus definitions alone. Machine-verifiable.
IMPORTED-PROVEN	A theorem proven in the external literature, cited here as a precedent; its Z-Spin realization is a separate claim.
DERIVED	Follows from the Z-Spin action plus PROVEN inputs; zero free parameters beyond A = 35/437.
DERIVED-CONDITIONAL	Derived, conditional on a stated upstream assumption (here the identifications A1–A3 of §5).
DERIVED-CANDIDATE	An algebraic identity holds, but its physical applicability rests on assumptions facing a known structural obstruction (here A1 vs the (2,3,6) blocks).
VERIFIED	Numerically confirmed against an independent computation or observation after all model-specific premises have been implemented. No claim in this paper is at this level: the CAMB result is a consistency check (OBSERVATIONALLY CONSISTENT), not VERIFIED, because it assumes standard cold-dark-matter transfer.
HYPOTHESIS-strong	Physically motivated and structurally well-supported; derivation chain incomplete.
BOOTSTRAP-HYPOTHESIS	An explicit, self-consistent ansatz (here an action) proposed for testing; its central claim is not yet derived and is gated by registered open problems.
OPEN	A sharply stated, falsifiable question that current corpus and external tools cannot settle. A registered asset locating the DERIVED/hypothetical boundary.
NO-GO	A proven impossibility for a specified construction; disciplines what the claim cannot be.
RETRACTED	A claim made in a prior version, now withdrawn with documented reason (here the v1.0 "one theorem closes both").
NON-CLAIM	Explicitly not asserted; documented to prevent overclaim.

§1. Introduction

The heights of the CMB acoustic peaks — in particular the third peak relative to the second — measure how strongly gravitational potential wells on acoustic scales were sustained, rather than allowed to decay under radiation pressure, through the radiation-to-matter transition. In ΛCDM this sustaining is performed by a cold, pressureless, non-free-streaming component: dark matter with $w \approx 0$ that clusters at recombination.

ZS-A18 (v1.5, frozen) analyzed whether the Z-Spin massless Goldstone field θ of the broken $U(1)_Z$ can perform this role and proved it cannot. A smooth massless scalar has, when time-dependent, $w = +1$ (kination), and as a static global-monopole gradient $w = -1/3$; it never has $w = 0$. This is the massless dichotomy. Subsequent free-exploration sessions confirmed and strengthened it: integrating out the heavy radial mode of the polar field still gives $w \in [1/3, 1]$ and $c_s^2 \in [1/3, 1]$, so no smooth-field configuration reaches $w = c_s^2 = 0$. ZS-A18 therefore stands as a NO-GO and is not modified by the present work.

Three object-based routes were then explored and each met a wall. (i) Planck-mass vortex cores: cold ($w = 0$) but, counted as relics, they over-close the universe by $\sim 10^{17}$–10^{28}. (ii) Current-carrying vortons: the dimensional reduction of a string to a cold worldline requires a conserved current, but ZS-S10's $j = 1/2$ core structure is a holonomy/BPS (Kraus-operator) object, not a Dirac zero mode, so the Jackiw–Rossi/Witten current carrier is absent and would require a new field; moreover the Z-Spin string tension is Planckian, placing any vorton far above the $\sim 10^9$–10^{12} GeV vorton-dark-matter window of Brandenberger–Carter–Davis–Trodden, hence over-closing. (iii) An intermediate $\sim 8.5\times10^8$ GeV formation scale would fix a Planck-mass relic abundance, but the corpus derives no such scale and any A,Q expression hitting it is rejected by anti-numerology.

The unifying lesson is that the obstruction is the *premise*, not the physics: every route assumed dark matter is a collection of objects. ZS-F2 already defines CDM differently — as a *geometric boundary tension, not a particle species* — with $\Omega_{cdm} = F(TI)/Q^2 = 32/121$. ZS-A18's NO-GO arose precisely because ZS-A1/A18 then identified that boundary tension with the massless Goldstone gradient. This paper takes ZS-F2 at its word and supplies the missing dynamics: a conserved geometric dust.

Principal contribution. (1) The over-closure/formation-scale/carrier obstructions are artifacts of the object ontology and are removed by a particle-free geometric dust (§3, §4). (2) The face count 32 is a candidate equivariant projection-class rank in the $Q^2 = 121$ channel algebra (§5, §6); but as Theorem A19.NG2 (§8) shows, a scalar-clock action sees this rank only through a removable multiplier normalization, so it does not by itself fix the cold fraction — its physical role requires relocation to a boundary Noether charge or operator-valued dynamics, which is open. (3) The residual is not a single node: the boundary energy matching (BRL-3), the channel-trace condition (BT-C), and the ZS-F23 modular-weight condition (Condition C) are distinct — Theorem A19.NG1 proves BT-C and Condition C are not the same trace — and several further gates (the $X\otimes Q$ channel, a rank-6 baryon projector, conserved-current matching, a relativistic-to-dust freeze-out, a global Z-anchor foliation, the ε-Halo overlap) are independent (§9.6). (4) This is therefore an explicit action ansatz with a registered self-no-go, reducing the third-peak CDM question to a small, ordered set of action-level and operator-algebraic gates rather than to one theorem.

§2. Locked Inputs from the Corpus

Geometric constants (LOCKED): **A** = $35/437 \approx 0.080092$, **Q** = 11, and the register decomposition (Z, X, Y) = (2, 3, 6) with $Q = 2 + 3 + 6$. No quantity in this paper introduces any parameter beyond these.

Dark-matter fraction (ZS-F2). The Boundary Mode Theorem registers $\Omega_{cdm} = F(TI)/Q^2 = 32/121$, with $F(TI) = 32$ the face count of the truncated icosahedron and $Q^2 = 121$, and explicitly characterizes CDM as a geometric boundary tension rather than a particle species. ZS-F2 supplies the fraction but not the stress tensor $T_{\mu\nu} = \rho\, u_\mu u_\nu$.

Logical algebra (ZS-Q11, ZS-F23). The single-cell Z-Spin logical algebra is the finite Type I von Neumann algebra $A_{ZS} \cong M_3(\mathbb{C}) \oplus \mathbb{C} \oplus M_5(\mathbb{C})$ (dimension 9 + 1 + 25 = 35, the numerator of **A**), with abelian center $\mathbb{C}^3$ carrying the three sector projections and equilibrium trace weights $p_{eq} = (3, 2, 6)/11$ for (X, Z, Y).

Emergence dictionary (ZS-F23, OPEN). ZS-F23 fixes the Type II crossed-product trace normalization geometrically; its residual is **Condition C** (O-F19.6 Step 1′): the gravitational de Sitter maximum-entropy state, coarse-grained to the Z-Spin sectors, carries the weights (3, 2, 6)/11. Condition C is registered as a genuine OPEN. We will show the present dark-matter bridge is its channel-space analogue.

Truncated icosahedron (TI). V = 60, E = 90, F = 32 (12 pentagons + 20 hexagons), full icosahedral symmetry I_h with $|I_h| = 120 = Q^2 - 1$; Euler characteristic V − E + F = 2.

Standing NO-GO (ZS-A18 v1.5, frozen). The massless Goldstone gradient is not recombination CDM. The present carrier is disjoint from this scope; ZS-A18 is unmodified (**NC-A19.2**).

§3. The Geometric-Dust Ontology

We replace the carrier ontology "CDM = collection of objects" with "CDM = conserved geometric dust from a constraint." The motivation is that a pressureless dust stress tensor can arise purely from a geometric constraint or a pullback action, with its energy density fixed by a constraint coefficient rather than by a relic abundance — so no over-closure calculation applies.

Theorem A19.1 (Existence of Particle-Free Geometric Dust) — IMPORTED-PROVEN

The following is an existence statement borrowed from the external literature: a pressureless dust can arise from a geometric constraint or pullback action, independently of any particle relic. Its Z-Spin realization — that the corpus action actually produces such a dust with the correct normalization — is a separate claim, registered below as HYPOTHESIS-strong/OPEN.

Each of the following established constructions yields an exact pressureless dust ($w = 0$, $c_s^2 = 0$, $\rho \propto a^{-3}$) without any particle number, rest mass, or formation temperature:

(A) Pullback perfect fluid. With three comoving scalar maps X^I and the identically conserved current $J^\mu = (1/3!)\, \varepsilon^{\mu\nu\rho\sigma} \varepsilon_{IJK}\, \partial_\nu X^I\, \partial_\rho X^J\, \partial_\sigma X^K$, proper density $b = (-J^\mu J_\mu)^{1/2}$, and action $S = \int \sqrt{-g}\, F(b)$ with energy density $\rho = -F$, the pressure is the convention-consistent $p = F - b F_b$ (equivalently $p = b \rho_b - \rho$). Choosing the equation-of-state input $F(b) = -\mu b$ gives $\rho = \mu b$ and $p = 0$, hence $T_{\mu\nu} = \rho\, u_\mu u_\nu$, $w = 0$, $c_s^2 = 0$; in FLRW $b \propto a^{-3}$ so $\rho \propto a^{-3}$. The linear form $F(b) = -\mu b$ is an independent physical input that selects the dust equation of state — it is not forced by the formalism — and the coefficient μ is the energy per conserved fluid element.

(B) Mimetic constraint. Imposing $g^{\mu\nu}\, \partial_\mu T\, \partial_\nu T = -1$ via a Lagrange multiplier Λ turns the conformal gravitational mode into pressureless dust with $\rho \propto a^{-3}$, $w = 0$, $c_s^2 = 0$.

(C) Projectable integration constant. With a space-independent lapse $N = N(t)$ the Hamiltonian constraint becomes a single global equation, leaving an integration-constant component $T_{\mu\nu} = \rho_{int}\, n_\mu n_\nu$ with $\rho_{int} \propto a^{-3}$ — pressureless dust with no matter field.

Consequence and scope. In every case the dust density is set by a constraint coefficient (μ, Λ, or an integration constant), not by a relic abundance, so the Kibble-density × Planck-mass over-closure estimate has no object to which it applies; the over-closure, formation-scale, and current-carrier obstructions are artifacts of the object ontology and do not arise here. The Goldstone field remains exactly massless, so the ZS-S3 theorem and the ZS-A1 ε-Halo are untouched: the dust is an additional component. Three caveats are explicit. (i) These are three *alternative precedents* from external gravity/fluid theory, not a single corpus theorem; each requires its own action structure (extra scalar maps, a mimetic constraint, or projectable lapse). (ii) The dust equation of state is fixed only after choosing $F(b) = -\mu b$ (or the analogous mimetic/projectable input). (iii) That the Z-Spin action itself produces this dust, and that μ is fixed by ZS-F2 rather than tuned, are the OPEN gates BTD-1 through BTD-4 of §13; mimetic and projectable routes additionally modify the gravitational sector and risk conflict with the ZS-M6 sector independence, which the pullback route avoids (§4).

§4. The X-Pullback Realization

Among the three mechanisms, the pullback perfect fluid is the most Z-Spin-native and the safest. It adds matter fields (the maps X^I) rather than gravitational scalar modes, so it preserves the existing diffeomorphism covariance and the ZS-M6 sector-independence $L_{XY} = 0$; the mimetic and projectable routes instead modify the gravitational sector and risk conflict with these structures (and the mimetic dust is prone to caustic and gradient instabilities).

Identification (HYPOTHESIS-strong). The X-sector carries three spatial register slots, dim(X) = 3. The pullback fluid requires exactly three comoving material coordinates X^I, $I = 1, 2, 3$. We identify the X-sector slots with these material coordinates, so that the ZS-F2 Y→X boundary transmission appears as a conserved X-space material current. This is structurally natural but changes the X-slots from purely kinematic degrees of freedom to dynamical fields and is therefore registered as HYPOTHESIS-strong; an action-level derivation is gate BTD-1 of §13.

Role of the Z-anchor (NON-CLAIM NC-A19.4). In this ontology the Z-anchor does not carry the CDM mass. It is the singular locus of the pullback map — where the Jacobian degenerates, $\det(\partial_\mu X^I) = 0$ — i.e. a defect/endpoint of the dust flow. The smooth boundary dust is the recombination CDM; the Z-anchor singularity is the locator of structure formation; late-time accretion supplies the SMBH mass; the massless Goldstone exterior is the separate late-time ε-Halo. This realizes the corpus principle "topology sets location; accretion sets mass" (ZS-F1 NC-2): the Z-anchor anchors the geometric flow rather than constituting it.

Version-Conflict Note (O-A19.6): the ε-Halo overlap

A genuine cross-paper tension must be flagged. If the pullback dust carries the full $\Omega_{cdm} = 32/121$ and clusters from recombination to today, it also clusters gravitationally in present-day galaxies. ZS-A1 separately attributes the entire galactic-rotation excess to the massless Goldstone ε-Halo gradient. Maintaining both as independent energy components would double-count gravity, $g_{total} = g_{baryon} + g_{dust} + g_{\varepsilon\text{-halo}}$. The naive "both retained as distinct components" does not resolve this. Three mutually exclusive resolutions are admissible, and selecting one is required: **(A)** the pullback dust replaces the ε-Halo dark-matter interpretation, with galactic halos re-derived as dust; **(B)** a transformation theorem identifies the linear cosmological dust and the nonlinear ε-Halo dressing as the same energy in different regimes, so the two are never summed; or **(C)** the ε-Halo is reinterpreted as the metric response of the dust rather than an independent mass-energy. This is registered as OPEN gate O-A19.6.

§5. The Equivariant Boundary-Rank–Trace Theorem

Theorem A19.1 fixes the equation of state but leaves the dust coefficient μ free. We now address the normalization $\Omega\_cdm = 32/121$. The claim under examination is that 32 is the rank of a candidate projection in the $Q^2 = 121$ channel algebra, whose normalized trace is exactly 32/121; as §5 itself concludes and §8 sharpens, this projection is a candidate rather than a canonical one, and its rank does not by itself fix a physical density. The representation theory below is verified independently in Appendix A.

Register representation (A1). Let $A_5 \cong I$ be the icosahedral rotation group acting on the 12 vertices of the icosahedron with stabilizer C_5. Deleting the constant vector from the 12-dimensional permutation representation gives $V_{11} = \mathbb{C}^{12} \ominus 1$, with character $\chi = (11, -1, -1, 1, 1)$ on classes (1A, 2A, 3A, 5A, 5B) and decomposition $V_{11} = 3 \oplus 3' \oplus 5$. We assume the Q = 11 register Hilbert space H_Q carries V_{11}.

Channel algebra (A3). The operator (channel) space is $K\_Q = H\_Q \otimes H\_Q^* \cong \mathrm{End}(V_{11}) \cong M_{11}(\mathbb{C})$, of dimension $Q^2 = 121$; its traceless part has dimension $120 = |I\_h|$, matching the corpus relation $Q^2 - 1 = 120$. As an A_5-module, $\mathrm{End}(V_{11}) = 3\cdot 1 \oplus 6\cdot 3 \oplus 6\cdot 3' \oplus 8\cdot 4 \oplus 10\cdot 5$.

Boundary module (A2). The 12 pentagonal faces form the orbit A_5/C_5 and the 20 hexagonal faces the orbit A_5/C_3, so the truncated-icosahedron face module is $F\_TI = \mathbb{C}[A_5/C_5] \oplus \mathbb{C}[A_5/C_3]$, with character $\chi\_F = (32, 0, 2, 2, 2)$ and decomposition $F\_TI = 2(1 \oplus 3 \oplus 3' \oplus 4 \oplus 5)$, dim = 32.

Since every A_5-irreducible occurs in $\mathrm{End}(V_{11})$ with multiplicity at least its multiplicity in F_TI ($3 \geq 2$, $6 \geq 2$, $6 \geq 2$, $8 \geq 2$, $10 \geq 2$), there exists an A_5-equivariant embedding $F\_TI \to \mathrm{End}(V_{11})$, with orthogonal projector P_BT onto the image of rank 32. Because $\mathrm{End}(V_{11})$ contains repeated irreducibles, this embedding is not unique: ∃ P_BT is proven, ∃! P_BT is not. P_BT is therefore a rank-32 projection candidate, not yet a canonical projection; canonical selection (gate O-A19.2) requires constructing P_BT as a spectral projector $1\_\Delta(\Delta\_transfer)$ of the Y→X transfer operator or a vertex–face incidence map, not merely matching characters.

Theorem A19.2 (Equivariant Rank–Trace) — algebraic identity DERIVED-CONDITIONAL on A1–A3; cosmological use DERIVED-CANDIDATE

With the canonical normalized trace $\tau\_Q(\cdot) = (1/Q^2)\,\mathrm{Tr}$ on the channel algebra, $\tau\_Q(P\_BT) = \mathrm{rank}\, P\_BT / Q^2 = 32/121$.

$$F(TI) = 32 \implies \mathrm{rank}\; PBT = 32 \implies \tau\boldsymbol{Q}(PBT) = 32/\boldsymbol{Q}^2 = 32/121$$

No equipartition assumption enters this *algebraic* identity: that every rank-one mode carries equal weight is the *definition* of the canonical normalized trace together with its unitary invariance, not a thermal or area-weighted input. This normalized channel trace on $\mathrm{End}(V_{11})$ is related to, but — by Theorem A19.NG1 — not identical to, the canonical trace ZS-F23 derives for the finite Z-Spin algebra A_ZS: the latter gives sector weights (3, 2, 6)/11, whereas the unique faithful unital embedding $A\_ZS \to M_{11}$ gives

(3, 3, 5)/11. The two are distinct traces on distinct algebras and must not be conflated. The identity here is, moreover, purely algebraic. Its *physical* use — that the cosmological dark-matter fraction equals 32/121, not merely that τ_Q(P_BT) = 32/121 — additionally requires that the physical state be the tracial (maximally mixed) state ρ* = I/121 and that the energy operator be degenerate on the boundary modes; these are Condition BT-C (§7). For a generic state ρ, Q_BT(ρ) = Tr(ρ P_BT) ≠ 32/121, and the energy fraction Ω_BT = Tr(ρ H P_BT)/Tr(ρ H) departs further still.

Table 1. Representation data (verified, Appendix A).

Object	A_5-character	Decomposition	dim
$V_{11} = \mathbb{C}^{12} \ominus 1$	(11, −1, −1, 1, 1)	$3 \oplus 3' \oplus 5$	11
$\mathrm{End}(V_{11}) = M_{11}$	(121, 1, 1, 1, 1)	$3\cdot 1 \oplus 6\cdot 3 \oplus 6\cdot 3' \oplus 8\cdot 4 \oplus 10\cdot 5$	121
F_TI (face module)	(32, 0, 2, 2, 2)	$2(1 \oplus 3 \oplus 3' \oplus 4 \oplus 5)$	32
P_BT (projector)	—	image of F_TI in $\mathrm{End}(V_{11})$	32

§6. What the 32-Charge Is and Is Not

Lemma A19.3a (Spatial-Topological NO-GO) — PROVEN

The boundary surface of the truncated icosahedron is topologically a sphere, $\partial(\mathrm{TI}) \cong S^2$, and $H^2(S^2, \mathbb{Z}) \cong \mathbb{Z}$ has a single generator. Subdividing one hexagonal face into smaller cells raises F above 32 without changing the topology. Hence F = 32 is not a spatial topological invariant; the statement "one topological charge per face, total 32" is false in ordinary topology.

Lemma A19.3b (Metric-Tension NO-GO) — PROVEN

The 32 faces are not congruent: 12 are regular pentagons (unit-edge area ≈ 1.7205) and 20 are regular hexagons (area ≈ 2.5981). A per-area (metric tension) weighting therefore does not reproduce the integer count: the hexagons alone carry an area fraction ≈ 0.716, not 20/32. Consequently the boundary charge cannot be a literal metric tension; it must be a combinatorial, quantized per-face charge. This disciplines ZS-F2's "boundary tension" language.

Conclusion. 32 is correctly read as an equivariant projection-class rank — a K_0-type quantity invariant under basis choice and unitary deformation — equal to dim F_TI = dim $C_2(\mathrm{TI}; \mathbb{C})$. It is the rank of P_BT in $\mathrm{End}(V_{11})$, and its canonical trace is 32/121 (§5). The phrase "topological face charge" should be replaced by "equivariant boundary-rank / projection-class charge."

§7. Condition BT-C and Its Relation to ZS-F23 Condition C

Theorem A19.2 establishes an *algebraic* identity, τ_Q(P_BT) = 32/121. The *physical* statement Ω_cdm = 32/121 requires that the cosmological state actually realize the canonical trace on the channel algebra. For a general state ρ the boundary weight is Q_BT(ρ) = Tr(ρ P_BT), and the energy fraction Ω_BT = Tr(ρ H P_BT)/Tr(ρ H); these equal 32/121 only if ρ is tracial and the boundary-mode energies are degenerate.

Condition BT-C (Boundary Tracial-State Matching) — OPEN

The Z-Spin gravitational/dust equilibrium state ω_grav, restricted to the channel algebra, equals the canonical normalized trace τ_Q; equivalently, the boundary transfer channel E_BT is primitive and unital (so its unique stationary state is maximally mixed) and the physical energy operator is degenerate on the boundary modes. Under BT-C, Q_BT = ω_grav(P_BT) = τ_Q(P_BT) = 32/121, and with the pullback action of §4, ρ_BT $\propto a^{-3}$ with p = 0, $c\_s^2 = 0$.

Theorem A19.NG1 (Trace-Incompatibility of A_ZS in M_{11}) — PROVEN NO-GO

A faithful unital *-representation of A_ZS = $M_3(\mathbb{C}) \oplus \mathbb{C} \oplus M_5(\mathbb{C})$ on the eleven-dimensional register requires multiplicities (m_3, m_1, m_5) with $3m_3 + m_1 + 5m_5 = 11$ and all $m_i \geq 1$; the unique solution is *$(m_3, m_1, m_5) = (1, 3, 1)$*, giving central-projection ranks (3, 3, 5) in M_{11} and hence M_{11} normalized-matrix-trace weights *(3, 3, 5)/11*. These are not the ZS-F23 equilibrium weights (3, 2, 6)/11 (the Z and Y slots differ). Therefore no single M_{11} normalized matrix trace can simultaneously induce the ZS-F23 sector weights and serve as the canonical trace giving τ_Q(P_BT) = 32/121.

Retraction. The v1.0 claim that "one gravitational-equilibrium-equals-canonical-trace theorem closes both the gravitational-entropy normalization (ZS-F23) and the dark-matter abundance" is **RETRACTED**. The two remain emergence-dictionary problems of the same kind — a physical state realizing a canonical trace — but they are not the same trace on the same algebra: the 32/121 lives in the 121-dimensional channel (operator) space with its Hilbert–Schmidt trace, while the (3, 2, 6)/11 are modular weights on the 11-dimensional register; NG1 shows the naive single-matrix-trace identification fails.

Repair paths (registered, not yet selected). (A) Keep the two traces distinct — the channel trace (32/121) and the modular trace ((3,2,6)/11) belong to different states/algebras; then the unification claim is simply dropped. (B) Matrix amplification: realizing center ranks proportional to (3, 2, 6) requires multiplicities $(m_3, m_1, m_5) = (5, 10, 6)$, i.e. an amplified algebra M_{55} with ranks (15, 10, 30) = 5·(3, 2, 6); this is mathematically available but demands a derivation of why the Q = 11 channel is amplified to 5Q = 55. (C) A non-tracial weighted density matrix on M_{11} can produce (3, 2, 6), but at the cost of the canonical-trace/equipartition/zero-fitted-parameter logic that underpins the 32/121 result. None of (A)–(C) is derived; this is the elevated form of gate O-A19.4.

Closed shortcut. One might hope the i-tetration contraction $|f'(z^*)| = 0.892 < 1$, which has a unique fixed point, supplies channel primitivity and hence equipartition. ZS-F23 §8 states explicitly that the i-tetration fixed point is orthogonal to the modular-trace content; this route is therefore closed by the corpus itself, and primitivity remains to be established independently.

§8. The Boundary-Rank-Locked Z-Clock Dust Mechanism

The §5–§7 route fixes the equation of state only up to the free coefficient μ, reaches the fraction 32/121 only through the canonical-trace condition BT-C, and runs into the A_5/(2,3,6) tension (O-A19.5) and the trace incompatibility (Theorem A19.NG1). This section presents a more corpus-native action-level route that targets the free μ and the A_5 tension and, in addition, writes an h-independent absolute number for the physical cold-dark-matter density. We call it Boundary-Rank-Locked Z-Clock Dust (BRL-ZCD). It combines three pieces — a rank-32 seam projector, a constrained Z-clock dust action, and a baryon-anchored boundary-matching term:

$$SBRL = SZS + Sclock + Smatch$$

Its overall status is BOOTSTRAP-HYPOTHESIS: an explicit, falsifiable action ansatz. The dust equation of state it produces is IMPORTED-PROVEN ordinary clock dust, but — as Theorem A19.NG2 (§8.2a) proves — the rank-32 projector is absorbed by a multiplier redefinition in this scalar action, so it does not by itself fix the cold fraction; the 32:6 ratio is imposed by S_match rather than derived; and the derivation of S_clock and S_match from the existing Z-Spin action remains OPEN. The honest residual is the gate inventory G1–G8 of §9.6, not a single node.

§8.1 The XQ−1 seam projector (replacing the A_5 route)

ZS-F2 supplies a second, independent route to the integer 32 that does not use the deleted-icosahedral representation V_{11}: the Z_2 gauge projection $XQ - 1 = 3\cdot 11 - 1 = 32 = F(TI)$. The Z-sector (dim 2) splits under Z_2 into one physical (Z_2-even, β_0) mode and one gauge (Z_2-odd) mode (ZS-S1 §5.2, PROVEN), and the cold channel does not couple to the single Z_2-odd gauge direction. We realize this as an explicit projector on the X⊗Q channel space. With the X-democratic direction $|s_X\rangle = (1,1,1)/\sqrt{3}$ and the Z-odd projector $P_Z^- = \frac{1}{2}(I_Z - J_Z)$,

$$P_c = I_{XQ} - P_X(0) \otimes P_Z^- , \quad \operatorname{rank} P_c = XQ - 1 = 33 - 1 = 32$$

Because this projector is built from the X-democratic and Z-odd directions rather than from $V_{11} = 3 \oplus 3' \oplus 5$, the three v1.1 obstructions — dim(**Z**) = 2 not an A_5-subrepresentation, the A_5/(2,3,6) non-commutation, and the Theorem A19.NG1 trace incompatibility — are not obstructions to this construction. The rank 32 is cross-checked against the independent face count F(TI) = 32 (§5). The uniqueness of the candidate null direction is PROVEN under the $S_3 \times Z_2$ representation (§9.2), but its realization as a physical X⊗Q channel and the physical relevance of its rank both remain OPEN — by Theorem A19.NG2 the rank is absorbed in the bulk scalar action. The status of gate BRL-2 is therefore OPEN, not DERIVED-CONDITIONAL; what the seam route buys is only that it sidesteps the A_5/(2,3,6) tension, not that it makes the rank physical.

§8.2 Theorem A19.3 (Constrained Z-clock dust): EOS IMPORTED-PROVEN, rank NO-GO (A19.NG2), Z-Spin realization OPEN

Promote the Z-sector "time-point" to a continuum clock field T_Z(x) and impose a unit-timelike constraint through a non-propagating Lagrange multiplier λ_c, projected onto the cold channel by P_c:

$$S_{clock} = -\tfrac{1}{2} \int d^4x \sqrt{-g}\ \ \mathrm{Tr}_{XQ}[\, \lambda_c P_c \,(g^{\mu\nu} \partial_\mu T_Z \partial_\nu T_Z + 1)\,]$$

Variation in λ_c gives $g^{\mu\nu} \partial_\mu T_Z \partial_\nu T_Z = -1$, so $u_\mu := \partial_\mu T_Z$ is unit timelike. Variation in T_Z gives $\nabla_\mu[\mathrm{Tr}(P_c)\, \lambda_c u^\mu] = 0$; since $\mathrm{Tr}(P_c) = 32$, defining $\rho_c := 32\, \lambda_c$ yields the conserved current $\nabla_\mu(\rho_c u^\mu) = 0$ — but this very step is where the rank is absorbed (Theorem A19.NG2, §8.2a). The metric variation on the constraint surface gives exactly

$$T_{\mu\nu}(c) = \rho_c u_\mu u_\nu , \quad p_c = 0 , \quad w_c = 0 , \quad c_{s,c}^2 = 0 , \quad \pi_c = 0 , \quad \rho_c \propto a^{-3}$$

The implication S_clock ⟹ $T_{\mu\nu} = \rho u_\mu u_\nu$ is PROVEN/IMPORTED-PROVEN (the Brown–Kuchař and mimetic clock/reference-field constructions). The Z-Spin content — that T_Z is the *existing* Z-sector reference clock rather than a newly added scalar — is HYPOTHESIS-strong (gate BRL-1). Crucially, T_Z is not the A18 Goldstone θ: constraining θ to be unit-timelike would change the theory A18 analyzed and could clash with the ZS-A1 angular halo phase, so the massless exterior Goldstone θ, the dust clock T_Z, and the density multiplier λ_c are kept distinct. If BRL-1 fails, the construction reduces to adding a new scalar — the generic mimetic-dust pattern — which is the explicit failure mode.

§8.2a Theorem A19.NG2 (Scalar-clock rank-absorption) — PROVEN NO-GO

The construction of §8.2 contains a decisive obstruction, identified in external review and recorded here. Because T_Z and λ_c are ordinary scalars, they act as the identity on the channel space, so the trace in S_{clock} factorizes:

$$Tr_{XQ}[\ \lambda_c P_c \mathscr{C}(T_Z)\ ] = \lambda_c \mathscr{C}(T_Z) \cdot Tr\, P_c = 32\, \lambda_c \mathscr{C}(T_Z)$$

with $\mathscr{C}(T_Z) = g^{\mu\nu}\partial_\mu T_Z\, \partial_\nu T_Z + 1$. Hence $S_{clock} = -16 \int\sqrt{(-g)}\, \lambda_c\, \mathscr{C}(T_Z)$, and the redefinition $\tilde{\lambda}_c = 32\, \lambda_c$ returns exactly the ordinary single-clock (mimetic) dust action $-\tfrac{1}{2}\int\sqrt{(-g)}\, \tilde{\lambda}_c\, \mathscr{C}(T_Z)$. The projector enters only through its rank, and that rank is removed by a redefinition of the multiplier. **Theorem A19.NG2 (PROVEN).** For a scalar clock T_Z and scalar multiplier λ_c, every action of the form $Tr[P\, \lambda_c\, \mathscr{C}(T_Z)]$ depends on the finite-rank projector P only through rank P, and that rank is removable by $\lambda_c \to \lambda_c/\text{rank } P$. Therefore no projector rank can determine the physical dust density or select a cold channel in this construction; any rank gives the same theory.

Consequence. The phrase "the cold rank is fixed by the rank-32 seam projector" is not valid for the bulk scalar action; what §8.2 actually exhibits is a conventional constrained dust action multiplied by the trace of a proposed rank-32 projector, with the 32 absorbed into the definition $\rho_c = 32\, \lambda_c$. The rank can be a physical observable only if it is relocated out of the bulk scalar action. Two repair directions are registered, both OPEN: **(R1) operator-valued dynamics —** promote λ_c or T_Z to a channel-operator-valued field with non-commuting P_c-dynamics, or construct 32 genuinely independent clock/current modes (this adds degrees of freedom and must be done without spoiling the locked sector structure); **(R2) boundary Noether charge —** the conservative repair: remove P_c from the bulk dust action entirely and let rank 32 appear only as a quantized per-face boundary Noether charge (gate G1, §9.6). Neither repair is carried out here; A19.NG2 stands as a standing no-go that any future rank-locking claim must clear.

§8.3 Baryon-anchored boundary matching (the abundance, with no μ)

ZS-F0 already carries a Gibbons–Hawking–York boundary term, a Robin boundary condition $n^\mu\partial_\mu\Phi = \mathbf{A}\Phi K$ (Theorem 8.1), and a BV-BFV functor in which the seam J is a boundary constraint operator, not a bulk symmetry (Theorem 8.7, $\|[J,L]\| \approx 2.94 \neq 0$). It is therefore corpus-native to fix the abundance by a boundary matching term rather than by a bulk potential. With the baryon rank $r_b = XZ = 3\cdot 2 = 6$ and the cold rank $r_c = XQ - 1 = 32$, on the post-confinement hypersurface Σ_{conf} where the baryon rest-mass current is defined,

$$S_{match} = \int_{\Sigma\_conf} d^3x\, \sqrt{h}\ \ \chi\, [\ (1/32)\, n_\mu n_\nu T_{\mu\nu}(c) - (1/6)\, n_\mu n_\nu T_{\mu\nu}(b)\ ]$$

Variation in the boundary multiplier χ enforces $\rho_c/32 = \rho_b/6$, i.e. the per-mode canonical energies of the two matter channels born at the same seam are equal, so

$$\rho_c / \rho_b = 32 / 6 = 16/3$$

Because both currents are separately conserved and dilute as a^{-3}, this ratio is time-independent. This is a boundary gluing condition, not a fit of μ to data — indeed μ never appears. Its status is HYPOTHESIS-strong, and three caveats are explicit. First, the same boundary form $S = \int\sqrt{h}\, \chi(\rho_c/r_c - \rho_b/r_b)$ imposes $\rho_c/\rho_b = r_c/r_b$ for any (r_c, r_b), so the term encodes the ratio 32:6 rather than deriving it — the honest reading is that it conditionally yields 16/3 under the postulated lock. Second, a stress-energy matching $n^\mu n^\nu T_{\mu\nu}$ is not the right invariant across Σ_{conf}, where the baryonic sector carries large thermal energy and is not yet the late-time pressureless rest-mass fluid; the matching should be written on conserved currents, $32^{-1} n_\mu J_c^\mu = 6^{-1} n_\mu J_B^\mu$ with $\nabla_\mu J = 0$, and even then a per-charge energy equality must be proven separately (gate G4). Third, a candidate rank-32 cold projector is proposed but no rank-6 baryon projector P_b ($P_b^2 = P_b$, rank 6, selecting the baryon channel) is constructed, so the 32:6 symmetry is one-sided (gate G3). The claim that the ZS-F0 BV-BFV boundary variation actually produces per-mode energy equality (gate BRL-3) is not proven.

§8.4 Theorem A19.4 (h-free absolute CDM density) — CONDITIONAL NUMERICAL CONSEQUENCE

The boundary lock fixes the ratio; the absolute scale follows from the corpus baryon asymmetry without using h. This is a conditional numerical consequence, not a derivation: it inherits the boundary lock (BRL-3, imposed), the conserved-current matching and per-charge energy equality (G4), and the external η_{10}–ω_b coefficient, in addition to the ZS-U3 value of η_B. ZS-U3/ZS-F5 derive $\eta_B = (Y/Q)^{35} = (6/11)^{35} = 6.117\times10^{-10}$ (the exponent 35 = numerator of **A** = dim A_ZS), matching Planck's $(6.12 \pm 0.04)\times10^{-10}$. The standard BBN relation $\eta_{10} = 273.9\ \omega_b$ (with T_{CMB} = 2.72548 K, FIRAS) then fixes the physical baryon density, and the boundary ratio fixes the cold density:

$$\omega b = \eta_{10} / 273.9 = 0.022334\ , \quad \omega c = (32/6)\ \omega b = 0.119112$$

This is reported at two tiers. **Tier 1 (parameter-free ratio):** conditional on the boundary lock, $\omega_c/\omega_b = 32/6 = 16/3 = 5.333$, against the Planck ratio 5.364 ± 0.065 — a -0.48σ agreement, with no η_B, no BBN coefficient, and no h. **Tier 2 (absolute, h-free):** $\omega_c = 0.119112$, against Planck $\omega_c = 0.1200 \pm 0.001$ — a -0.89σ agreement, additionally conditional on the U3 value of η_B and the external η_{10}–ω_b coefficient. The honest status of this number is therefore DERIVED-CONDITIONAL on U3 + BRL-3 + the conserved-current matching, not on U3 alone. It nonetheless replaces ZS-F2's $\omega_c = (32/121)h^2$, which re-inserts the Planck-measured h and is circular at the level of the physical density.

Anti-numerology (AN-A19.2, EXECUTED, PASS). The integers 32 (= XQ − 1 = F(TI)) and 6 (= XZ) pre-exist in ZS-F2, and $\eta_B = (6/11)^{35}$ pre-exists in ZS-U3, all fixed before this paper, so $\omega_c = 0.119112$ is a post-hoc output, not a fit. η_B and ω_b are not independent observations (ZS-F5 Independence Warning §10.7); they are therefore not counted as two successes. The coefficient 273.9 is an external BBN input via T_{CMB}, and μ is never used. No new fitted parameter enters.

§8.5 Perturbations and adiabaticity

Because the mechanism is exact pressureless dust, its linear perturbations obey the standard cold-dark-matter Boltzmann system (Ma–Bertschinger): in synchronous gauge $\delta_c' = -\theta_c - \frac{1}{2}h'$ and $\theta_c' = -H\theta_c$, reducing in the comoving gauge to $\delta_c' = -\frac{1}{2}h'$. Applying the boundary lock $\rho_c/32 = \rho_b/6$ *locally* (not only to the mean density) linearizes to $\delta_c = \delta_b$; with a single-clock inflationary origin giving $\delta_b = (3/4)\delta_\gamma$, this yields $\delta_c = (3/4)\delta_\gamma$, hence the isocurvature mode vanishes, $S_{c\gamma} = \delta_c - (3/4)\delta_\gamma = 0$ (adiabatic; gate BRL-4). After the constraint is released the cold component free-streams geodesically while baryons stay coupled to photons, reproducing standard acoustic evolution; with no source term the component is passive, so the third-peak potential wells at $k_3 \approx 0.058\ \mathrm{Mpc}^{-1}$ are sustained without a smeared defect hump. The implication $\delta_c = \delta_b \Rightarrow S_{c\gamma} = 0$ is itself correct (PROVEN linear algebra), but its status as a closed gate is OPEN, not DERIVED-CONDITIONAL, for three reasons: the density contrast is gauge-dependent, so the equality must be stated as the gauge-invariant entropy perturbation $S_{cb} = \delta\rho_c/(\rho_c+p_c) - \delta\rho_b/(\rho_b+p_b) = 0$; $p_b \approx 0$ does not hold immediately after confinement; and a single-clock adiabatic mode does not by itself remove an independent dust-density integration constant of the multiplier λ_c, which must be locked to the inflaton mode by a separate action-level constraint. The full CLASS/CAMB confirmation is gate BRL-5.

§8.6 What BRL-ZCD bypasses, and what it retains

Table 2. v1.1 bottlenecks and their BRL-ZCD handling.

v1.1 bottleneck	BRL-ZCD handling
Three X-pullback fields with free μ	single constrained Z-clock T_Z; density is the multiplier λ_c (no μ)
A_5 rank-32 projector (O-A19.5 tension)	XQ−1 seam projector inside the locked (2,3,6) decomposition
Theorem A19.NG1 trace incompatibility	not on the abundance path, but the rank itself is absorbed in the bulk action (A19.NG2): the seam route avoids NG1 without yet making the rank physical
Condition BT-C (maximally-mixed state)	relocated, not removed: the boundary gluing imposes the ratio, and its per-mode energy equality is itself a BT-C-type equipartition condition (§9.4)
32/121 as today's fraction (radiation equipartition)	32:6 ratio + h-free $\eta_B \to \omega_b \to \omega_c$ absolute normalization
Goldstone mass / 10^9 GeV scale / Planck relic	none used; dust is multiplier-sourced, $m_\theta = 0$ preserved

The A_5 rank–trace identity (§5) and Theorem A19.NG1 (§7) are not discarded: the former remains an independent geometric cross-check that the integer is 32 (F(TI) = XQ − 1), and the latter remains a PROVEN no-go that disciplines any future attempt to unify the abundance with the ZS-F23 weights. They are simply off the abundance critical path in this route.

§8.7 Version-conflict note (F2 reinterpretation) and the BRL gate set

Cross-paper honesty requires flagging a partial reinterpretation of ZS-F2. A distinction must be drawn that Theorem A19.NG2 forces: the **mode-count ratio** $r_c : r_b = 32 : 6$ is a corpus-derived count (DERIVED — it is $g_{eff}(CDM) = 32$ and XZ = 6), whereas the **physical energy-density ratio** $\rho_c : \rho_b = 32 : 6$ is OPEN: in v1.4 it is imposed by the boundary term S_{match} (which can impose any ratio), and the route by which a count becomes a density is exactly the closure program of §10. We down-weight and reinterpret F2's final step — carrying $\Omega_{cdm} = 32/121$ to today's dust fraction via radiation-era Stefan–Boltzmann equipartition — because ZS-U6 notes that this equipartition does not transfer directly to the matter era. This is a registered version-conflict, not a contradiction of F2's integer content.

Table 3. BRL-ZCD gate set — v1.4 status (the v1.2 "DERIVED-CONDITIONAL / single link" reading is superseded by §9 and the gate inventory G1–G8).

Gate	Requirement	Status (v1.4)
BRL-1	T_Z is an existing Z-sector reference clock, not a new propagating scalar; worldline proper time ≠ global scalar clock.	HYP-strong/OPEN
BRL-2	$P_g = P_X^{(0)} \otimes P_Z^-$ is the unique gauge-null mode AND its rank is physical; rank absorbed in the bulk action (A19.NG2), X⊗Q not derived.	OPEN

Gate	Requirement	Status (v1.4)
BRL-3	The ZS-F0 BV-BFV boundary variation enforces $\rho_c/32 = \rho_b/6$ ($\neq$ BT-C $\neq$ F23 Condition C per NG1).	OPEN
BRL-4	The local lock gives the gauge-invariant $S_{cb} = 0$ with no residual isocurvature.	OPEN
BRL-5	Direct CLASS/CAMB implementation passes TT, TE, EE, lensing, and P(k).	OPEN
BRL-6	Cosmological dust and the galactic ε-Halo do not double-count (nonlinear matching).	OPEN

Overall status. BOOTSTRAP-HYPOTHESIS: an explicit, falsifiable action ansatz. The dust equation of state is IMPORTED-PROVEN ordinary clock dust; Theorem A19.NG2 shows the bulk rank-32 lock is absorbed; the physical Z-Spin origin $S_{ZS} \Longrightarrow S_{BRL}$ is unproven; and the residual is not a single link but the gate set G1–G8 (§9.6). §10 presents the most conservative route by which these gates might close.

§9. Gate Testing, Status, and Registered Obstructions

v1.3 tested the §8 gates by direct computation. v1.4 records an external technical review that found a decisive obstruction those tests missed (Theorem A19.NG2, §8.2a), and recalibrates every status accordingly. The honest summary is a sharp contraction: the dust equation of state is rigorous, a new self-no-go bounds the mechanism, and the abundance and origin remain an explicit action ansatz gated by several distinct open problems.

§9.1 BRL-5 (acoustic peaks): COMPUTED / OBSERVATIONALLY CONSISTENT

Because BRL-ZCD posits that the linear dust transfer is identical to standard cold dark matter, the only number BRL-5 can test is $\omega_c = 0.119112$. A CAMB Boltzmann integration, holding h, n_s, A_s, τ, and the neutrino sector at the Planck 2018 values and varying only (ω_b, ω_c), gives the third acoustic peak +0.10% from the Planck best fit, with controls at $\omega_c = 0.10$ and 0.14 shifted by ±2.4%.

Table 4. CAMB third-peak computation (other parameters fixed at Planck 2018; standard-CDM transfer assumed).

Model	ω_c	Third peak (ℓ, $\mathscr{D}$ / μK^2)	vs Planck
Planck 2018	0.12000	(813, 2541.3)	—
BRL-ZCD value	0.119112	(814, 2543.9)	+0.10% (−0.74σ)
control (low)	0.100	(834, 2603.6)	+2.45%
control (high)	0.140	(796, 2482.2)	−2.32%

Status: **COMPUTED / OBSERVATIONALLY CONSISTENT**, not VERIFIED. Three honest limits. (i) The computation assumes standard cold-dark-matter transfer — exactly the thing BRL-ZCD must derive — so it confirms that $\omega_c = 0.119112$ is allowed by a standard transfer, not that S_{BRL} produces that transfer. Since 0.119112 differs from Planck's 0.1200 by only 0.7%, a near-identical peak is expected. (ii) The peak-region $\Delta\chi^2$ for the controls holds all other cosmological and nuisance parameters fixed; it is a peak-shape comparison, not a parameter-likelihood exclusion, and must not be read as one. (iii) The original BRL-5 gate requires the full TT, TE, EE, lensing, and P(k) likelihood; only the TT third peak is computed here. BRL-5 therefore remains OPEN as a derivation gate.

§9.2 BRL-2 (seam mode): a uniqueness result, but the gate is OPEN

One sub-question of BRL-2 — which democratic direction is selected — does have a clean answer. The Z_2-odd factor $P\_Z^- = ½(I\_Z − J\_Z)$ has rank 1 (ZS-S1 §5.2, PROVEN); the X-democratic factor is the unique invariant of the octahedral S_3 permuting the three coordinate axes (ZS-M6 §5.5), the Reynolds projector $P\_X^{(0)} = |S_3|^{-1} \Sigma\_\sigma U\_\sigma = |s\_X\rangle\langle s\_X|$ with $|s\_X\rangle = (1,1,1)/\sqrt{3}$, and the continuous SO(3) has no invariant direction (NO-GO). So if the construction is granted, the gauge-null mode is unique. **But two obstructions keep the gate OPEN.** First, by Theorem A19.NG2 the rank of this projector is absorbed in the bulk scalar action, so even a unique P_c does not fix the cold fraction — uniqueness of the projector and physical relevance of its rank are different questions, and only the first is addressed. Second, the 33-dimensional space X⊗Q is a tensor product, whereas the corpus register is a direct sum H_Q = H_Z ⊕ H_X ⊕ H_Y; the counting identity XQ = 33 does not by itself supply a physical tensor-product channel K_c ≅ H_X ⊗ H_Q (gate G2). Status HYPOTHESIS-strong/OPEN.

§9.3 BRL-4 (adiabaticity): correct implication, open gate

The implication $\delta\_c = \delta\_b \Longrightarrow S\_c\gamma = \delta\_c - (3/4)\delta\_\gamma = 0$ is PROVEN linear algebra (given $\delta\_b = (3/4)\delta\_\gamma$). But, as §8.5 details, its status as a closed gate is OPEN: the density contrast is gauge-dependent and must be replaced by the gauge-invariant entropy perturbation S_cb = 0; p_b ≈ 0 fails right after confinement; and the multiplier λ_c can carry an independent dust-density integration constant not removed by a single-clock adiabatic mode. The implication is right; the premises are not yet established.

§9.4 Equipartition (Theorem A19.5): IMPORTED-PROVEN precedent, application OPEN

The boundary equality ρ_c/32 = ρ_b/6 is an equipartition statement. Maximum-entropy / Stefan–Boltzmann arguments are a useful precedent but do not close it. **Theorem A19.5 (downgraded to IMPORTED-PROVEN precedent).** The unconstrained maximum-entropy state is maximally mixed (Jaynes), but the physical state at fixed energy is the Gibbs state $e^{-\beta H}/Z$; a rank-to-energy identity ρ_c : ρ_b = 32 : 6 holds only in the high-temperature limit where the modes are relativistically thermalized at a common temperature, i.e. only under mode-energy degeneracy H_c ≃ H_b — which is precisely the BT-C open question, not something Jaynes removes. **Three distinct sub-gates.** (a) the H-degeneracy itself; (b) a relativistic-equipartition → freeze-out → pressureless-dust transition that preserves 32:6 (the carrier must end with w = 0, not the w = 1/3 of the thermal modes); (c) the baryon thermal history — electroweak/QCD transitions, baryon–antibaryon annihilation, entropy transfer, and nucleosynthesis — between the high-T baryon degrees of freedom and today's rest-mass density, so that the high-T rank ratio is not simply the late-time ω_c : ω_b. None of these is supplied by the thermodynamic precedent.

§9.5 BRL-1: the dust clock is HYPOTHESIS-strong/OPEN; a narrowed NO-GO

In a flat FLRW background, a clock T(τ) is unit-timelike iff T = ±τ + const. The proper time of a comoving congruence satisfies this tautologically, which is why the Z-anchor (Bogomolnyi vortex-core) proper time is a natural candidate for T_Z, distinct from the Goldstone θ. But a worldline proper time is a global scalar clock field only under further conditions, none of which is established: the congruence must be hypersurface-orthogonal (irrotational, $u\_{[\mu} \nabla\_\nu u\_{\rho]} = 0$) for $u\_\mu = \partial\_\mu T\_Z$ to hold; it must fill spacetime as a smooth global foliation without worldline crossings or caustics; a sparse, SMBH-like anchor distribution must coarse-grain to a smooth cold fluid; and the congruence must already exist at recombination. Status HYPOTHESIS-strong/OPEN (gate G7), not DERIVED-CONDITIONAL.

Theorem A19.6 (i-tetration clock, PROVEN but NARROWED). The ZS-F10 internal time $\nu = (A/\pi)\ln(\tau/t_P)$ is not itself a normalized proper-time clock: $g^{\mu\nu}\partial_\mu\nu\, \partial_\nu\nu = -(A/\pi)^2/\tau^2 \neq -1$. This rules out ν as the dust clock. It does *not* exclude the i-tetration structure as a source of a proper-time clock: the inverse map $\tau(\nu) = t_P\, e^{\pi\nu/A}$ recovers a unit-timelike proper time. The NO-GO is ν-specific; the NON-CLAIM is bounded accordingly.

§9.6 Honest residual: distinct gates, not one node

The v1.3 claim that three routes converge on a single emergence node is withdrawn. It contradicts the paper's own Theorem A19.NG1, which proves the channel normalized trace (Condition BT-C, weights (3,3,5)/11 on M_{11}) and the ZS-F23 modular weights (Condition C, (3,2,6)/11) are distinct traces on distinct algebras. The boundary energy matching BRL-3, the channel-trace condition BT-C, and the modular-weight condition Condition C are therefore three different conditions, and several further gates are independent. The honest residual is the following set.

Table 5. Honest gate inventory after v1.4 (replacing the v1.3 single-node claim).

Gate	Open problem	Status
G1	Relocate rank 32 out of the bulk scalar action (absorbed, A19.NG2) into a quantized boundary Noether charge, or operator-valued clock dynamics.	OPEN
G2	Derive the tensor-product channel $K_c \cong H_X \otimes H_Q$ from the direct-sum register $H_Z \oplus H_X \oplus H_Y$.	OPEN
G3	Construct a rank-6 baryon projector P_b ($P_b^2=P_b$, rank 6) selecting the baryon channel.	OPEN
G4	Boundary matching on conserved currents (not stress-energy), plus a per-charge energy equality.	OPEN
G5	Relativistic-equipartition → freeze-out → pressureless dust preserving 32:6 (with baryon thermal history).	OPEN
G6a	BT-C: the physical state realizes the channel normalized trace.	OPEN
G6b	F23 Condition C: the gravitational equilibrium state realizes (3,2,6)/11 — distinct from G6a per NG1.	OPEN
G7	Global irrotational Z-anchor foliation realizing a single dust clock field T_Z.	OPEN
G8	ε-Halo nonlinear double-counting (BRL-6); full TT/TE/EE/lensing/P(k) likelihood (BRL-5).	OPEN

Net. $S_{ZS} \Rightarrow S_{BRL}$ does not close, and the residual is not one node but at least 6–8 distinct open problems. What is solid is narrow and worth stating: the geometric dust equation of state is IMPORTED-PROVEN, the integers 32 and 6 pre-exist in the corpus, the A18 NO-GO is intact, and Theorem A19.NG2 now bounds the mechanism by ruling out a bulk rank-lock. The overall status of BRL-ZCD is a BOOTSTRAP-HYPOTHESIS: an explicit, falsifiable action ansatz with a registered self-no-go, not a HYPOTHESIS-strong near-closure.

§10. A Boundary-Charge Closure Program (ZHCS) for the Gate Set

Theorem A19.NG2 forbids closing the abundance by strengthening the bulk scalar-clock action: any rank placed there is absorbed into a multiplier. This section presents the most conservative route by which the gate set G1–G8 of §9.6 might nonetheless close — not by adding an ansatz, but by relocating the rank out of the bulk entirely and onto a conserved boundary canonical charge, with the bulk supplying only ordinary Brown–Kuchař dust. We call the construction Z-Seam Harmonic mass-Charge Splitting (ZHCS). It is a closure program, not a closure: the rank-ratio result below is PROVEN conditional on two registered theorems, ZHCS-1 (a finite, immediately computable matrix fact) and ZHCS-2 (the real action-level bottleneck).

The key idea, and why it evades A19.NG2. Let a single boundary canonical momentum p live on a connected graph whose nodes are the 32 cold faces and the 6 baryon faces. If the boundary variation forces p to be graph-harmonic ($d_\Gamma p = 0$) and the graph is connected, then p is constant across all 38 nodes, $p = p_\star 1_{38}$. The two matter charges are projections of this one parent charge: $Q_c = 1^T P_c p = 32 p_\star$ and $Q_b = 1^T P_b p = 6 p_\star$. A common rescaling $p_\star \to \alpha p_\star$ multiplies both, so the ratio $Q_c/Q_b = 32/6$ is invariant. A19.NG2 removes the absolute scale (exactly this α); it does not remove the ratio of two projections of a common parent charge. The number 32:6 is never written into the action — it is the rank of two fixed face modules read off a connected boundary.

§10.1 The long list and the conservative choice

Seven repair routes were enumerated. (L1) operator-valued clock and (L2) thirty-two independent bulk clocks both evade A19.NG2 but add many degrees of freedom, breaking Z-Spin minimality. (L3) thermal equipartition then dust freeze-out is the unproven transition already registered as gate G5. (L4) a projectable integration-constant dust modifies the gravitational Hamiltonian constraint. (L5) a quantized per-face flux re-introduces an occupancy assumption. The two adopted are (L6) a common parent BFV charge that branches into a 32-dimensional and a 6-dimensional module, and (L7) a connected dual-face graph whose harmonic charge is automatically equipartitioned; the bulk uses the externally established Brown–Kuchař dust. L6+L7 add no new field beyond the existing boundary phase space and no new number beyond the two fixed face-module ranks.

§10.2 The parent matter face module (closes G3 conditionally)

The cold module is the ZS-F2 truncated-icosahedron face module $E_c = C^2(TI)$, $\dim E_c = 32$ (the same $32 = F(TI) = XQ - 1$ used throughout). The baryon module is the six square faces of the X-sector truncated octahedron, $E_b = C^2_\square(tO) \cong C^2(\text{cube})$, $\dim E_b = 6 = XZ$. Their direct sum is the parent matter face space

$$Em = Ec \oplus Eb , \quad \dim Em = 32 + 6 = 38$$

On this direct sum the projectors $P_c = \mathrm{diag}(I_{32}, 0)$ and $P_b = \mathrm{diag}(0, I_6)$ are canonical (not a non-unique embedding), with $P_c + P_b = I_{38}$, rank $P_c = 32$, rank $P_b = 6$. This supplies the rank-6 baryon projector that v1.4 lacked (gate G3): its status is DERIVED-CONDITIONAL, the condition being the identification of $E_c \oplus E_b$ as the physical matter boundary phase space. Note that the construction is a direct sum, sidestepping gate G2 (the unproven $X \otimes Q$ tensor product) rather than resolving it.

§10.3 The Z-Spin-mediated seam graph and ZHCS-1 (now DERIVED)

Declaring $E_c \oplus E_b$ does not by itself equalize the two blocks' charge density; a transfer structure connecting them is required. The corpus carries the X–Z–Y transfer matrices C_{XZ}, K_Z, C_{ZY} with no direct X↔Y block: $L_{XY} \equiv 0$ is PROVEN (ZS-F1, ZS-S1, ZS-M6 §7A), so the sectors communicate only through Z-mediation, and that mediation is the PROVEN rank-1 β_0-selected channel $C_{ZX} \approx \kappa|z_0\rangle\langle r_X|$, $C_{ZY} \approx \kappa|r_Y\rangle\langle z_0|$ with $\kappa^2 = A/Q \neq 0$ (ZS-F9 §6.6, ZS-M6 §2.2). With vertex-to-face incidences R_b, R_c the face-level cross-transfer $W_{bc} = R_b^T C_{XZ} K_Z C_{ZY} R_c$ is therefore rank-1, $W_{bc} = u v^T$. The seam graph Γ_m has the 6 baryon (cube) faces, the 32 cold (TI) faces, the polyhedral face-adjacency inside each block, and cross-edges $w_{ij} = |(W_{bc})_{ij}|^2$; the decisive question is connectivity, ZHCS-1: rank $L_\Gamma = 37 \Leftrightarrow \beta_0(\Gamma_m) = 1$.

Status: DERIVED. This is now settled by explicit computation on the actual polyhedra, from PROVEN inputs and with no tuning. (i) The cold block is the truncated-icosahedron face graph: built from the icosahedron (12 pentagons of degree 5, 20 hexagons of degree 6; a pentagon and a hexagon adjacent iff the icosahedron vertex lies on the icosahedron face, two hexagons adjacent iff the faces share an edge), its Laplacian has rank 31 — connected. (ii) The baryon block is the cube face graph (the 6 = F(cube) = XZ channel, ZS-F2 §11.4); its Laplacian has rank 5 — connected. Both are face-adjacency graphs of convex polyhedra, hence connected by Steinitz's theorem independently of the computation. (iii) The cross-block has full support: the Euler null mode of the TI vertex-face incidence is, by ZS-F2 Lemma 4.5 (PROVEN), $v_p = -2v_h$ uniformly on all 12 pentagons and v_h on all 20 hexagons with $(\Sigma_f v)^2 = 4/17$ — nonzero on every cold face — and the analogous tO null mode is nonzero on all 6 squares, so the rank-1 $W_{bc} = u v^T$ has u, v nowhere zero and the cross-graph is complete bipartite. Assembling the 38-node graph gives rank $L_\Gamma = 37$ ($\ker L_\Gamma = \mathrm{span}\{1_{38}\}$); the result is robust, holding already for a single cross-edge because both blocks are internally connected.

The conditionality is honestly bounded. Connectivity uses the cube face-adjacency for the baryon block (the natural reading, since the 6 squares are the cube's 6 = XZ faces). The companion computation also records the failure mode for contrast: if the 6 squares were instead taken as mutually non-adjacent truncated-octahedron faces and the cross-support reached only one O_h dipole pair (two faces), the four remaining squares would be isolated and the rank would drop to 33. Connectivity therefore rests on two PROVEN facts — polyhedral block-connectivity and the nonzero, full-support β_0 mediation (Lemma 4.5) — and on the cube identification of the baryon faces; under these it is DERIVED.

§10.4 The harmonic boundary charge and ZHCS-2 (now DERIVED-CONDITIONAL)

The boundary action carries no energy multiplier; it carries only a harmonicity constraint on the parent charge through a boundary Lagrange field χ:

$$S_\Sigma^{harm} = \int_{\Sigma^*} d^3x \sqrt{h}\ \langle \chi , d_\Gamma p\rangle \quad \Rightarrow (\text{vary } \chi) \quad d_\Gamma p = 0$$

For a connected graph, standard graph cohomology gives $\ker d_\Gamma = H^0(\Gamma_m) = \mathrm{span}\{1_{38}\}$, hence $p = p_\star(x)\, 1_{38}$. This is what replaces equipartition: it uses no thermal equilibrium, no Stefan–Boltzmann law, no ZS-F23 canonical trace, and no 32:6 input — only connected boundary gluing. The theorem to be established is that this constraint descends from ZS-F0:

$$\text{ZHCS-2:} \quad \delta S_{F0}^{boundary} \quad \Rightarrow \quad d_\Gamma p = 0$$

Status: DERIVED-CONDITIONAL. Three corpus and imported results assemble the scaffold. (a) ZS-F0 §8.2 gives the GHY term and the Robin boundary condition $n^\mu\partial_\mu\Phi = A\Phi K$ (Thm 8.1, PROVEN), and §8.3–§8.4 the BV-BFV functor B_Z with the seam J a boundary constraint operator, not a bulk symmetry ($\|[J,L]\| \approx 2.94 \neq 0$, Thm 8.7, PROVEN). (b) The modified quantum master equation of the BV-BFV formalism, $(\hbar^2\Delta\_\mathcal{V} + \Omega\_\partial\Sigma)\psi\_\Sigma = 0$ with $\{\Delta\_\mathcal{V}, \Omega\_\partial\Sigma\} = 0$ (IMPORTED-PROVEN, Cattaneo–Moshayedi–Wernli 2018), states that the boundary state is annihilated by the boundary BFV operator $\Omega\_\partial\Sigma$ — the abstract closedness condition. (c) The BRST Hodge decomposition (IMPORTED-PROVEN, Malik 2000) identifies the physical states with the harmonic representatives, harmonic = closed ∩ co-closed; on 0-cochains the discrete Hodge Laplacian is exactly the graph Laplacian $L\_\Gamma = d\_\Gamma^T d\_\Gamma$, whose harmonic space (ker L_Γ) is span$\{1_{38}\}$ for the connected Γ_m of §10.3. Composing, the boundary charge is a BFV-harmonic 0-cochain, $p \in \ker L\_\Gamma$, i.e. $d\_\Gamma\, p = 0$. **The single residual conditional** is the explicit identification of the boundary BFV operator with the graph coboundary, $\Omega\_\partial\Sigma\,|_0$-cochain $\cong d\_\Gamma$ — i.e. writing the ZS-F0 boundary symplectic potential in the 38-face basis and confirming $\Omega\_\partial\Sigma$ acts as d_Γ with no residual independent zero mode. The BV-BFV machinery supplies a harmonic boundary state; this identification fixes that the relevant Hodge Laplacian is the seam-graph L_Γ. It is the one remaining gap, reduced from the v1.5 "real bottleneck, OPEN" to a single operator identification.

§10.5 The rank-ratio theorem (now DERIVED-CONDITIONAL)

Given ZHCS-1 (connectivity, DERIVED) and ZHCS-2 (harmonicity, DERIVED-CONDITIONAL), the cold and baryon charges $Q\_c = 1^T P\_c\, p$ and $Q\_b = 1^T P\_b\, p$ satisfy

$$Qc = 32\, p^\star\,, \quad Qb = 6\, p^\star\,, \quad Qc/Qb = 32/6 = 16/3$$

With ZHCS-1 now DERIVED (§10.3), this is PROVEN conditional on ZHCS-2 alone, and since ZHCS-2 is DERIVED-CONDITIONAL (§10.4), the ratio's net status is DERIVED-CONDITIONAL. It does not violate A19.NG2: the common normalization $p\_\star \to \alpha\, p\_\star$ acts on both charges, so while the absolute scale is (correctly) unfixed, the ratio is rescaling-invariant. The number 32:6 here is the rank of two fixed face modules of a connected boundary, not a multiplier ratio inserted into the action. This is the precise sense in which v2.0 turns the imposed v1.4 ratio into a conditional consequence — the conditional now being a single boundary-operator identification, not two open problems.

§10.6 Brown–Kuchař bulk dust and canonical branching (G1, G4, G7)

The bulk uses the full Brown–Kuchař reference dust, not a single mimetic scalar:

$$Sc\hat{}\{BK\} = -\tfrac{1}{2} \int d^4x \sqrt{(-g)}\;\; \rho c\; (\, g\hat{}\{\mu\nu\}\; U\mu\; U\nu + 1\,)\,, \quad U\mu = -\partial\mu\; TZ + WI\; \partial\mu\; X\hat{}I$$

with T_Z the Z-even reference clock, X^I (I = 1,2,3) the X-sector comoving labels, and W_I velocity potentials. The variation gives $T\_\{\mu\nu\} = \rho\_c\, U\_\mu\, U\_\nu$, $\nabla\_\mu(\rho\_c\, U^\mu) = 0$, $U^\mu \nabla\_\mu U\_\nu = 0$, hence $w\_c = c\_s^2 = \pi\_c = 0$ and $\rho\_c \propto a^{-3}$ (IMPORTED-PROVEN). This eases the v1.4 BRL-1 problem: T_Z and the three X^I together constitute the dust reference system, so no single worldline proper time need be promoted to a global scalar, and vorticity is admitted through W_I — the Z-anchor survives only as a Jacobian defect / locator, not as the whole congruence (gate G7). The rank now enters only through the boundary branching $S\_branch = \int\_\{\Sigma^*\} \sqrt{h}\, [\langle \Lambda\_c, \Pi\_c - P\_c\, p\rangle + \langle \Lambda\_b, \Pi\_b - P\_b\, p\rangle]$ with the clocks glued $T\_c| = T\_b| = T\_\star$; since $P\_c + P\_b = I$, the boundary symplectic potentials add to $\Theta\_c + \Theta\_b = \langle p, \delta T\_\star\rangle = \Theta\_\star$, so the branching conserves the symplectic charge and contains no 32 or 6 in the action (gates G1, G4). That this branching is the genuine canonical gluing of the ZS-F0 boundary to the Brown–Kuchař momentum is the registered theorem ZHCS-3 (OPEN).

§10.7 Adiabaticity from a common parent charge (G5-adjacent)

Because both species are projections of the one parent charge, a perturbation $p_\star = \bar{p}_\star(1 + \delta_\star)$ gives $\delta Q_c/\bar{Q}_c = \delta Q_b/\bar{Q}_b = \delta_\star$, i.e. the gauge-invariant curvature perturbations coincide, $\zeta_c = \zeta_b$, so the cold–baryon entropy perturbation $S_{cb} = 3(\zeta_c - \zeta_b) = 0$ — gauge-invariantly, repairing the gauge-dependent $\delta_c = \delta_b$ of v1.4. A single parent charge also leaves no independent dust-density integration mode. If the parent clock $T_\star$ is the single reheating clock, then $\zeta_c = \zeta_b = \zeta_\gamma$ and $S_{c\gamma} = 0$; that identification is the registered theorem ZHCS-4 (OPEN). The relativistic-to-dust freeze-out of gate G5 is removed from the critical path: the charge is harmonic by boundary gluing, not by thermal equipartition.

§10.8 The four minimal theorems and the honest verdict

Table 6. The four ZHCS theorems and their v2.0 status after the deep-exploration closure.

Theorem	Statement (what must be proven)	Status (v2.0)
ZHCS-1	The 38-node Z-seam face graph Γ_m is connected: rank $L_\Gamma = 37$ ($\beta_0 = 1$). Verified by building the TI face graph (32, connected) and cube face graph (6, connected) and computing the Laplacian rank; cross-block full support from Lemma 4.5.	DERIVED
ZHCS-2	The ZS-F0 boundary variation yields $d_\Gamma\, p = 0$. Scaffold = mQME ($\Omega_{\partial\Sigma}\psi=0$) + BRST Hodge + seam J boundary operator; residual = the identification $\Omega_{\partial\Sigma} \cong d_\Gamma$ on the 38-face basis.	DERIVED-COND
ZHCS-3	$\Pi_c = P_c\, p$ and $\Pi_b = P_b\, p$ glue symplectically to the Brown–Kuchař dust momentum and the baryon conserved-current momentum.	OPEN
ZHCS-4	A single parent clock perturbation exists, giving $\zeta_c = \zeta_b = \zeta_\gamma$ (no independent dust-density / isocurvature mode).	OPEN

Verdict. ZHCS reduces the eight gates G1–G8 to these four theorems: it relocates the rank to a boundary charge (G1), supplies the rank-6 projector by direct sum (G3), replaces thermal equipartition by connected-graph harmonicity (G5), and removes BT-C and ZS-F23 Condition C from the abundance critical path (G6a, G6b). In v2.0 the deep-exploration closes the first two: ZHCS-1 is DERIVED (the seam graph is connected, rank $L_\Gamma = 37$, computed on the actual truncated-icosahedron and cube face graphs, with full cross-support from the PROVEN Euler null mode of Lemma 4.5), and ZHCS-2 is DERIVED-CONDITIONAL (the BV-BFV mQME, the BRST Hodge decomposition, and the seam-J boundary operator give the harmonic-charge scaffold, leaving only the operator identification $\Omega_{\partial\Sigma} \cong d_\Gamma$). Consequently the rank-ratio theorem (§10.5) rises from PROVEN-CONDITIONAL on two open gates to **DERIVED-CONDITIONAL on the single ZHCS-2 identification**. The overall program status is correspondingly **DERIVED-CONDITIONAL** for the abundance ratio: the third-peak ratio 32:6 is no longer a numerological mode-count or an imposed boundary condition but the harmonic canonical charge of a connected BFV seam branching into two fixed face modules, evading A19.NG2 by common rescaling — conditional on the one operator identification (ZHCS-2) and, for the full perturbation spectrum, on ZHCS-3/4. This is reported as a conditional derivation, not an unconditional one; ZHCS-2's explicit face-basis variation and ZHCS-3/4 remain the registered residuals.

§11. Consistency Conditions and Registered Gaps

Beyond the OPEN bridge BT-C, several structural conditions must hold for the §5 construction to be the corpus's own. One of them is now a proven obstruction (Theorem A19.NG1) rather than a mere open question:

O-A19.4 (Algebra / trace identification — now sharpened by Theorem A19.NG1). The rank-32 projector lives in the 121-dimensional channel (operator) space, whereas the corpus logical algebra A_ZS $= M_3(\mathbb{C}) \oplus \mathbb{C} \oplus M_5(\mathbb{C})$ sits inside M_{11} with the unique faithful unital embedding giving center weights (3, 3, 5)/11. By Theorem A19.NG1 this is not the ZS-F23 (3, 2, 6)/11, so the two normalizations cannot be the same matrix trace. Reconciliation requires one of the repair paths of §7 (separate traces, M_{55} amplification, or a non-tracial weighted state), none derived.

O-A19.5 (A_5 vs the (2,3,6) sector blocks — stronger than a consistency note). The A_5-invariant subspace dimensions of $V_{11} = 3 \oplus 3' \oplus 5$ are the partial sums {0, 3, 5, 6, 8, 11}; the value 2 does not occur, so dim(**Z**) = 2 is not an A_5-subrepresentation. This is not merely an unproven identification: if A_5 is the full register symmetry then the Z-sector is not preserved, and if the Z-sector is an exact physical block then the full A_5 action does not commute with the (2, 3, 6) decomposition. The simultaneous use of the locked block structure (including L_XY = 0) and the rank-32 A_5 construction must therefore be re-examined. Admissible repairs: let A_5 act only on a separate face/channel module rather than on H_Q; let A_5 break to a subgroup at low energy producing 2 + 3 + 6; withdraw the V_{11} identification; or construct an explicit intertwiner between the sector projectors and the A_5 action. Pending this, the cosmological status of Theorem A19.2 is DERIVED-CANDIDATE rather than DERIVED-CONDITIONAL.

O-A19.2 (Canonicality). Because End(V_{11}) contains repeated irreducibles, the rank-32 embedding is not unique: ∃ P_BT is proven, ∃! P_BT is not. A canonical selection requires a physical principle (vertex–face incidence, Z-seam parity, the Y→X transfer spectral projector, or a boundary-channel fixed-point algebra).

O-A19.3 (Primitivity / dynamical equipartition). Selecting the canonical trace by hand re-imports equipartition. A dynamical derivation requires the boundary channel E_BT to be unital and primitive; a simple A_5-twirl is insufficient because the representation is reducible and the pentagon/hexagon sectors leave a w_5/w_6 freedom. Establishing primitivity (full Kraus algebra, trivial peripheral spectrum) is OPEN.

§12. Consistency with Observation

The geometric partition reproduces the Planck 2018 ΛCDM densities to within current uncertainties, and the matter and dark-energy slots close to unity:

Table 7. Geometric partition versus Planck 2018.

Component	Fraction	Value	Planck 2018	Status
Baryons Ω_b	6/121	0.0496	0.0493	PASS
Cold DM Ω_cdm	32/121	0.2645	0.264	PASS
Matter Ω_m	38/121	0.3140	0.315	PASS
Dark energy Ω_Λ	83/121	0.6860	0.685	PASS

With h = 0.6736, $\Omega_{cdm} h^2 = (32/121)(0.6736)^2 = 0.12000$, matching the measured physical CDM density $\omega_c = 0.120$. The matter and dark-energy slots satisfy (38 + 83)/121 = 1. No observational tension is introduced, and the ZS-A18 NO-GO is preserved: the present carrier is geometric dust, disjoint from the Goldstone gradient. ZS-A18's cold-fluid CAMB control run already showed that a component with exactly $w = 0$, $c_s^2 = 0$, $\omega_c = 0.120$ reproduces the third peak. That run becomes the actual leading-order transfer calculation for the boundary dust only once the full gate set is closed — not under BT-C alone, but with BT-C *and* the action-level origin (BTD-1–3), the normalization (BTD-4), adiabatic and passive initial conditions with negligible anisotropic stress (BTD-5: $S_{BT\gamma} = 0$, $\pi_{BT} \approx 0$, no continual source), and an explicit coupling to the metric-perturbation equations. Until then it remains a control calculation, not a derivation of the peak.

The route sharpens the cold-dark-matter comparison in a way that does not reuse the Planck-measured h. As of v2.0 the structural ratio is a DERIVED-CONDITIONAL output of the ZHCS harmonic boundary charge (§10), conditional on the single ZHCS-2 operator identification, rather than merely imposed: $\omega_c/\omega_b = 32/6 = 16/3 = 5.333$, against the Planck ratio 5.364 ± 0.065 (-0.48σ). At the absolute level, with the ZS-U3 baryon asymmetry $\eta_B = (6/11)^{35}$ fixing $\omega_b = 0.022334$ through the standard BBN relation, the numerical consequence is $\omega_c = (32/6)\ \omega_b = 0.119112$, against Planck $\omega_c = 0.1200 \pm 0.001$ (-0.89σ). Both are consistent with Planck 2018; neither uses h, removing the circularity of ZS-F2's $\omega_c = (32/121)h^2$. As emphasized in §8.4, η_B and ω_b are not independent observations and are not counted as two successes, and $\omega_c = 0.119112$ is a conditional output of pre-existing corpus quantities, not a fit.

§13. Falsification Gates

Table 8. Multilayered falsification gates for ZS-A19 (v1.5).

Gate	Layer	Falsification condition	Status
F-A19.1	Math (immediate)	A_5 representation theory shown wrong ($V_{11} \neq 3 \oplus 3' \oplus 5$, or $\chi_F \neq (32,0,2,2,2)$, or rank $P_{BT} \neq 32$).	PASSING
F-A19.2	Math	F_{TI} shown not to embed A_5-equivariantly in End(V_{11}) → rank–trace theorem void.	PASSING
F-A19.3	Theoretical (demotion)	Condition BT-C shown impossible (gravitational equilibrium state provably ≠ canonical trace) → §5 demoted to NON-CLAIM for cosmology.	OPEN
F-A19.4	Simulation/consistency	A boundary-dust transfer in CLASS/CAMB fails to reproduce the third peak at $\omega_c = 0.120$ → revision.	OPEN
F-A19.5	Cross-paper	Closing BT-C requires modifying A = 35/437 or Q = 11, or contradicts the ZS-A18 NO-GO → reject.	PASSING
F-A19.6	Observational	The partition 6/121, 32/121, 38/121, 83/121 shown inconsistent with updated cosmological data → reject.	PASSING
F-A19.7	Anti-numerology	The dust coefficient μ must be fitted to observation (no derivation from F2) → the 32/121 normalization is numerology; demote.	PASSING
F-A19.8	Cross-paper (NG1)	All three repair paths of §7 (separate traces, M_{55} amplification, weighted state) shown unavailable → the A_{ZS}/M_{11} normalizations are permanently incompatible and the F23 link is void.	OPEN

Gate	Layer	Falsification condition	Status
F-A19.9	Cross-paper (ε-Halo)	No resolution of O-A19.6 exists (the pullback dust and the ZS-A1 ε-Halo provably double-count galactic gravity) → the carrier conflicts with ZS-A1.	OPEN
F-A19.10	Observational (BRL)	A CLASS/CAMB run of the boundary-dust transfer with $\omega_c = 0.119112$ fails TT/TE/EE/lensing/P(k), or future data move ω_c outside $0.119112 \pm$ few σ → BRL-ZCD revised.	OPEN
F-A19.11	Theoretical (BRL)	T_Z is shown to be necessarily a new propagating scalar (BRL-1 fails) or the ZS-F0 boundary variation provably cannot yield $\rho_c/32 = \rho_b/6$ (BRL-3 fails) → BRL-ZCD reduces to generic mimetic dust and loses its corpus-native status.	OPEN
F-A19.12	Theoretical (clock)	The Z-anchor congruence is shown not to be unit-timelike or not irrotational at recombination scales, so $T_Z = \tau_{core}$ fails → the v1.3 clock identification (§9.5) is retracted; or the i-tetration time is shown to be unit-timelike after all, voiding Theorem A19.6.	OPEN
F-A19.13	Decisive (emergence)	A microscopic computation or measurement of the gravitational equilibrium state, coarse-grained to the Z-Spin sectors, gives weights ≠ rank-proportional (≠ 32:6 for the abundance, ≠ (3,2,6)/11 for ZS-F23) → the emergence dictionary fails and the abundance program is falsified (shared with F-F23.8).	OPEN (decisive)
F-A19.14	Decisive (rank)	No construction relocates the rank 32 into a physical observable — neither a quantized boundary Noether charge nor operator-valued clock dynamics makes the cold fraction depend on the rank (gate G1) → by Theorem A19.NG2 the rank-32 claim is physically empty and BRL-ZCD reduces to ordinary single-clock dust with an externally imposed 32:6 ratio.	OPEN (decisive)

Anti-numerology (AN-A19.1, EXECUTED, PASS). ZS-A19 introduces no parameter beyond (A, Q, (Z, X, Y)). The seemingly attractive instanton survival probability $P_{cand} = (A/2Q)\, e^{-\pi/A} \approx 3.4\times10^{-20}$, numerically close to the relic-dilution target $\approx 3.6\times10^{-20}$, is REJECTED: neither π/A as a bounce action nor $A/2Q$ as a fluctuation determinant is derived, and the expression was selected after seeing the target. The $\sim10^{9}$ GeV scale appearing in object routes is the Brandenberger–Carter–Davis vorton-dark-matter scale, used only as external context, never as a Z-Spin derivation. The normalization 32/121 is admissible only if μ is fixed by F2 without observational fitting (gate F-A19.7); fitting μ is the explicit failure mode.

§14. The Closure Program and Non-Claims

Two action-level routes are now on the table. The primary route is the geometric Z-clock dust of §8, whose gates BRL-1 through BRL-6 were tested in v1.3 and recalibrated in v1.4 (§9). The original X-pullback route of v1.0–v1.1, with the gates BTD-1 through BTD-6 below, is retained as a nonlinear completion and an independent cross-check. The honest picture is: the dust equation of state is IMPORTED-PROVEN; Theorem A19.NG2 rules out a bulk rank-lock, so the rank-32 claim must be relocated to a boundary charge; the 32:6 ratio is imposed in v1.4, not derived; equipartition is an imported precedent with an open application; the CAMB result is observationally consistent, not verified; and the residual is the set of 6–8 distinct gates G1–G8 of §9.6, not a single node — BT-C and the ZS-F23 modular weights (Condition C) being distinct per Theorem A19.NG1. The ZHCS program of §10 is the most conservative concrete route by which G1–G8 might close, reducing them to the four theorems ZHCS-1 through ZHCS-4; ZHCS-1 (graph connectivity) is an immediately computable finite gate and ZHCS-2 (the ZS-F0 boundary variation) is the genuine remaining bottleneck. Each gate is to be settled by action variation, a Noether/current computation, an external PROVEN theorem, or a finite matrix computation — never by fitting a number or a scale:

Table 9. Closure program (Z-Spin Boundary-Tension-to-Dust Theorem).

Gate	Requirement	Status
BTD-1	X-sector slots realized as action-level comoving maps $X^I : M \to T^3$.	HYP-strong
BTD-2	The conserved current J^μ emerges from F0/F2 boundary variation as a quantized per-face (projection-class) charge, not a metric tension.	OPEN
BTD-3	The boundary energy takes the pullback form $S = -\int \sqrt{-g}\, \mu\, (-J^2)^{1/2}$, giving $p = 0$, $c_s^2 = 0$, $\rho \propto a^{-3}$.	DERIVED-COND
BTD-4	$Q_{BT}/Q_{total} = 32/121$ from the face/projection structure (Theorem A19.2) with no observational fit of μ.	DERIVED-COND
BTD-5	Adiabatic, passive initial conditions $S_{BT\gamma} = 0$ from a single Z-Spin inflationary clock.	OPEN
BTD-6	Z-anchor identified with the Jacobian singularity $\det(\partial X) = 0$, linked to the late-time SMBH locator.	HYP-strong
NG1 / trace	Reconcile the A_{ZS}/M_{11} normalizations via one of the §7 repair paths (separate traces, M_{55} amplification, or weighted state).	NO-GO+OPEN
A_5-align	Make the A_5 (V_{11}) structure compatible with the (2,3,6) sector blocks and $L_{XY} = 0$ (intertwiner, subgroup breaking, or restricted action).	OPEN
O-A19.6	Resolve the ε-Halo overlap so the dust and the ZS-A1 Goldstone halo do not double-count galactic gravity.	OPEN
Boltzmann	Implement the boundary-dust perturbations (adiabatic IC, $\pi \approx 0$, metric coupling) in CLASS/CAMB and reproduce the C_ℓ and $P(k)$.	OPEN

Non-claims. NC-A19.1: ZS-A19 does not claim the third peak is solved; the physical bridge (BT-C) and the gate sets of Tables 3 and 9 are OPEN. NC-A19.2: ZS-A19 does not contradict ZS-A18; it proposes a different carrier outside the Goldstone/smooth-field scope of that NO-GO, which remains frozen at v1.5. NC-A19.3: the $\sim 10^9$ GeV vorton scale and the instanton probability are not used as derivations (rejected by anti-numerology). NC-A19.4: Z-anchors are not the CDM mass carriers; they are the locators (Jacobian singularities) of the dust flow. NC-A19.5: ZS-A19 does not claim a single theorem closes both the gravitational-entropy normalization and the dark-matter abundance — that v1.0 claim is RETRACTED by Theorem A19.NG1. NC-A19.6: the v1.1 pullback normalization μ is not derived from ZS-F2; the v1.2 BRL-ZCD route removes μ entirely but introduces instead the unproven identification of T_Z with an existing Z-sector degree (BRL-1) and the unproven boundary-energy equality (BRL-3). NC-A19.7: the number $\omega_c = 0.119112$ is a conditional numerical output (DERIVED-CONDITIONAL on ZS-U3's η_B, the imposed boundary lock, and the external BBN coefficient) — it is not an independent measurement and η_B and ω_b are not counted as two successes; the unresolved structure is the gate set G1–G8 (§9.6), which is not reducible to a single action-origin link.

§15. Conclusion

Every previously blocked route to the Z-Spin third peak assumed dark matter is a collection of objects, and each met an obstruction — over-closure, an underived ~10^9 GeV formation scale, or an absent current carrier — that this paper identifies as an artifact of that premise. Taking ZS-F2's definition of CDM as a geometric boundary tension literally, and supplying its dynamics through a conserved pullback dust, removes the equation-of-state, sound-speed, and over-closure problems simultaneously, with the Goldstone field left exactly massless and the ZS-A18 NO-GO intact.

The dark-matter fraction 32/121 is expressed as a machine-checked algebraic statement: the truncated-icosahedron face module embeds A_5-equivariantly in the $Q^2 = 121$ channel space as a rank-32 projection whose canonical normalized trace is exactly 32/121, with the "32 charge" disciplined as a projection-class rank rather than a spatial topological charge or a metric tension. Two limitations are explicit: that projection is a candidate, since every rank-32 projection shares the trace and the dynamics must still select the face-module image; and the algebraic identity becomes the physical fraction only under Condition BT-C. BT-C is of the same kind as ZS-F23 Condition C, but Theorem A19.NG1 shows they are not the same trace on the same algebra — the unique faithful unital embedding of A_ZS in M_{11} gives center weights (3, 3, 5)/11, not the ZS-F23 (3, 2, 6)/11 — so the v1.0 claim that one theorem closes both is withdrawn. What remains is a finite, ordered set of action-level and operator-algebraic gates (Table 9), led by BT-C but not reducible to it, together with the registered tensions $\dim(Z) = 2 \notin V_{11}$ and the A_ZS/M_{11} trace incompatibility, and the ε-Halo overlap with ZS-A1.

v1.2 added a more corpus-native action-level route, Boundary-Rank-Locked Z-Clock Dust: a constrained Z-clock dust action that gives exact pressureless dust, with a proposed rank-32 seam projector and a baryon-anchored boundary lock that, anchored to the corpus baryon asymmetry, yields h-free numbers $\omega_c = 0.119112$ (−0.89σ) and the ratio $\omega_c/\omega_b = 16/3$ (−0.48σ). v1.4 bounds this route rigorously. The dust equation of state is IMPORTED-PROVEN, but Theorem A19.NG2 shows the rank-32 projector is absorbed by a multiplier redefinition in the bulk scalar action, so the cold fraction is not fixed by the rank there; the 32:6 ratio is imposed by the boundary term S_match, which can impose any ratio; and the two numbers above are conditional consequences of the imposed lock plus external inputs, not predictions. The third-peak question is thereby converted not into one theorem but into a small, ordered set of distinct action-level and operator-algebraic gates (G1–G8, §9.6), with $S\_ZS \Rightarrow S\_BRL$ unproven.

In v1.4, an external technical review prompted a sharp recalibration. The headline correction is Theorem A19.NG2: a scalar-clock action sees the seam projector only through its trace, which a multiplier redefinition removes, so the central "rank-locked" claim does not hold in the bulk action and the rank must be relocated to a boundary Noether charge. The v1.3 statement that three routes converge on a single emergence node is withdrawn — by the paper's own Theorem A19.NG1 the channel trace and the ZS-F23 modular weights are distinct — and the abundance is gated by several distinct open problems (mode-energy degeneracy, a freeze-out preserving 32:6, the baryon thermal history, a rank-6 baryon projector, conserved-current matching) rather than one. The honest end state is that the geometric dust equation of state is rigorous and the integers pre-exist, while the abundance and the action-level origin remain an explicit ansatz with a registered self-no-go and a set of 6–8 falsifiable gates; the ZS-A18 NO-GO is intact and no new parameter is fitted. Adding A19.NG2 and recording these downgrades raises the honesty of the program, and locating that boundary precisely is the contribution.

v1.5 takes the next step the no-go demands: instead of repairing the bulk action, it relocates the rank to a conserved boundary canonical charge and asks the gates to close there. The ZHCS construction of §10 lets one parent charge live on a connected 38-face graph of the truncated-icosahedron cold faces and the truncated-octahedron baryon faces; if the ZS-F0 boundary variation makes that charge graph-harmonic, the ratio 32:6 emerges as the rescaling-invariant ratio of two projections of one parent charge — evading A19.NG2, which removes only the absolute scale — and the bulk is ordinary Brown–Kuchař dust, which dissolves the global-clock difficulty into a reference-dust system. This converts the eight gates into four sharply stated theorems, of which ZHCS-1 (connectivity) is an immediately computable finite-matrix fact and ZHCS-2 (that the ZS-F0 variation actually yields graph-harmonicity) is the single genuine action-level bottleneck. The rank-ratio theorem is proven conditional on those two; the program adds no new fitted number and keeps the ZS-A18 NO-GO intact. ZS-A19 thus ends not with a closure but with a precise, falsifiable, and partly computable map of exactly what remains: if the connectivity computation succeeds and the ZS-F0 boundary variation delivers harmonicity, the third-peak dark-matter ratio of the Z-Spin cosmology would, for the first time, stand above DERIVED-CONDITIONAL.

v2.0 performs that program. The connectivity computation succeeds: built explicitly from the icosahedron, the truncated-icosahedron face graph (12 pentagons, 20 hexagons) is connected and the cube face graph is connected, so the 38-node seam graph has Laplacian rank $L_\Gamma = 37$ — ZHCS-1 is DERIVED, with the cross-block shown to have full support by the PROVEN Euler null mode of the truncated-icosahedron vertex-face incidence ($(\Sigma v)^2 = 4/17$, nonzero on every face). The ZS-F0 boundary variation does not yet deliver harmonicity unconditionally, but it nearly does: the BV-BFV modified quantum master equation, the BRST Hodge decomposition, and the seam J as a boundary constraint operator together force the boundary charge to be a harmonic 0-cochain, leaving only the identification of the boundary BFV operator with the seam-graph coboundary — so ZHCS-2 is DERIVED-CONDITIONAL. The net effect is the advance the program was built for: the third-peak ratio 32:6 rises from PROVEN-CONDITIONAL on two open gates to DERIVED-CONDITIONAL on a single operator identification, with A19.NG2 respected throughout (the ratio is the rescaling-invariant quotient of two projections of one parent charge). What ZS-A19 does not claim is an unconditional derivation: ZHCS-2's explicit 38-face variation and the perturbation theorems ZHCS-3/4 are the registered residuals, and the dust equation of state, the integers, and the ZS-A18 NO-GO are exactly as before. The honest end state of v2.0 is a third-peak dark-matter ratio that is, for the first time, a conditional consequence of connected boundary geometry rather than an imposed number — one operator identification away from a full derivation.

Acknowledgements and Code Availability

This paper consolidates internal Z-Spin Collaboration deep-exploration notes (free-exploration sessions, June 2026) following ZS-A18 v1.5. All 60 checks are reproducible from two consolidated companion scripts (zs_a19_verify_v2_0.py and zhcs_closure_test.py) archived in the repository: the representation-theory, partition, equation-of-state, NG1, and anti-numerology checks; the CAMB acoustic-peak computation of §9.1 (optional `--camb`, else recorded reference values); the seam-mode, adiabaticity, maximum-entropy, and i-tetration tests of §9.2–§9.5; the four review-driven obstruction proofs (the A19.NG2 rank-absorption of §8.2a, the S_match arbitrary-ratio, the maximum-entropy energy/rank gap, and the i-tetration reparametrization); the four ZHCS structural checks of §10; and the v2.0 ZHCS-1 polyhedral closure computation — the truncated-icosahedron face graph built from the

icosahedron (32 nodes, Laplacian rank 31), the cube face graph (6 nodes, rank 5), the 38-node seam graph (rank $L_\Gamma = 37$), the Euler null mode of the TI vertex-face incidence (ZS-F2 Lemma 4.5: unique kernel, $v_p = -2v_h$, $(\Sigma v)^2 = 4/17$, nonzero on all 32 faces), and the recorded failure mode (isolated squares → rank 33). The external dust constructions, index theorems, vorton constraints, von Neumann embedding, maximum-entropy, BV-BFV / mQME, BRST-Hodge, and thermal-time results are cited in full in the References. This work used AI tools (Anthropic Claude) for verification and drafting; the author assumes full responsibility for all content.

Appendix A. Representation-Theory Verification

All character computations use the A_5 table on classes (1A, 2A, 3A, 5A, 5B) with sizes (1, 15, 20, 12, 12) and irreducible characters $1 = (1,1,1,1,1)$, $3 = (3,-1,0,\varphi,\varphi')$, $3' = (3,-1,0,\varphi',\varphi)$, $4 = (4,0,1,-1,-1)$, $5 = (5,1,-1,0,0)$, where $\varphi = (1+\sqrt{5})/2$, $\varphi' = (1-\sqrt{5})/2$; $\Sigma \dim^2 = 1+9+9+16+25 = 60$.

The 12-vertex permutation character is (12, 0, 0, 2, 2) (order-5 elements fix the two axial vertices; orders 2, 3 fix none); subtracting the trivial gives $\chi(V_{11}) = (11, -1, -1, 1, 1)$, with inner products $\langle V_{11}, 3\rangle = \langle V_{11}, 3'\rangle = \langle V_{11}, 5\rangle = 1$ and $\langle V_{11}, 1\rangle = \langle V_{11}, 4\rangle = 0$, so $V_{11} = 3 \oplus 3' \oplus 5$. Squaring gives $\chi(\mathrm{End}\ V_{11}) = (121, 1, 1, 1, 1)$, with multiplicities (3, 6, 6, 8, 10) on (1, 3, 3′, 4, 5) and traceless dimension $120 = |I_h|$.

The pentagon orbit A_5/C_5 has character (12, 0, 0, 2, 2) and the hexagon orbit A_5/C_3 has character (20, 0, 2, 0, 0); their sum $\chi(F_{TI}) = (32, 0, 2, 2, 2)$ has every irreducible inner product equal to 2, so $F_{TI} = 2(1 \oplus 3 \oplus 3' \oplus 4 \oplus 5)$, dim 32. Each multiplicity in F_{TI} (all 2) is at most the corresponding multiplicity in $\mathrm{End}(V_{11})$ (3, 6, 6, 8, 10), so an A_5-equivariant embedding exists and P_{BT} has rank 32; the canonical normalized trace gives $\tau_Q(P_{BT}) = 32/121$. For Theorem A19.NG1, the faithful unital embedding of $A_{ZS} = M_3 \oplus \mathbb{C} \oplus M_5$ on $\mathbb{C}^{11}$ has the unique multiplicity solution $(m_3, m_1, m_5) = (1, 3, 1)$ of $3m_3 + m_1 + 5m_5 = 11$, giving center ranks (3, 3, 5) and M_{11} trace weights $(3, 3, 5)/11 \neq (3, 2, 6)/11$; the minimal amplification realizing $5\cdot(3, 2, 6)$ is (5, 10, 6) on M_{55}. The pullback pressure is $p = F - bF_b$, vanishing for $F = -\mu b$.

For the v1.2 BRL-ZCD route: the seam projector $P_c = I_{XQ} - P_X^{(0)} \otimes P_Z^-$ has rank $XQ - 1 = 33 - 1 = 32$, matching F(TI) = 32 independently of the A_5 representation; the constrained Z-clock action gives (via the λ_c and T_Z variations) a conserved current $\nabla_\mu(\rho_c u^\mu) = 0$ with $\rho_c = 32\lambda_c$, hence $p_c = 0$, $c_s^2 = 0$, and $\rho_c \propto a^{-3}$ from FLRW continuity; the corpus baryon asymmetry $\eta_B = (6/11)^{35} = 6.117\times10^{-10}$ gives $\omega_b = \eta_{10}/273.9 = 0.022334$ and, through the boundary ratio 32/6, $\omega_c = 0.119112$ (-0.89σ from Planck $\omega_c = 0.1200 \pm 0.001$), with the parameter-free ratio $\omega_c/\omega_b = 16/3 = 5.333$ at -0.48σ. All 35 algebraic and arithmetic claims of this paper pass (machine-checked); none of the six action-level cosmological gates (BRL-1–6) is closed.

v1.3 added thirteen checks and v1.4 four more (52 total); the four v1.4 checks are obstruction proofs that recalibrate, not raise, the statuses. CAMB (camb 1.6.6): with $(\omega_b,\omega_c)=(0.022334,0.119112)$ the third peak is at $(\ell,\mathscr{D})=(814, 2543.9\ \mu K^2)$, +0.10% from the Planck best fit and -0.74σ from $\omega_c=0.1200\pm0.0012$, with controls $\omega_c=0.10/0.14$ shifted by +2.45%/−2.32%; this is a consistency check assuming standard cold-dark-matter transfer (TT third peak only; the control $\Delta\chi^2$ is a peak-shape comparison at fixed parameters, not a likelihood). BRL-2 (uniqueness sub-result): the S_3 Reynolds projector on $\mathbb{C}^3$ has rank 1 and equals $|s_X\rangle\langle s_X|$, $|s_X\rangle=(1,1,1)/\sqrt{3}$; the J_Z Z_2-odd projector has rank 1; SO(3) shares no invariant direction (invariant operator the rank-3 identity), so rank $P_c=33-1=32$. BRL-4 (implication): $\delta_c=\delta_b \Rightarrow S_{c\gamma}=0$ (exact). Equipartition: the unconstrained maximum-entropy state on $\mathbb{C}^d$ is maximally mixed ($S=\ln d$, $d=3,11$), but the fixed-energy state is Gibbs $e^{-\beta H}/Z$, so $\rho_c:\rho_b=32:6$ holds only in the high-T degenerate limit. BRL-1: $\nu=(A/\pi)\ln(\tau/t_P)$ has $g^{\mu\nu}\partial_\mu\nu\partial_\nu\nu=-(A/\pi)^2/\tau^2\neq-1$ (NO-GO), while proper time has -1 (unit-timelike), and $\tau(\nu)=t_P e^{\pi\nu/A}$ recovers it (so the NO-GO is ν-specific). v1.4 review-driven proofs: (i) $\mathrm{Tr}[\lambda_c P_c \mathscr{C}]=32\,\lambda_c \mathscr{C}$, removable by $\lambda_c\to\lambda_c/32$ (A19.NG2, rank absorbed); (ii) $\int\sqrt{h}\,\chi(\rho_c/a-\rho_b/b)$ imposes $\rho_c/\rho_b=a/b$ for any (a,b) (32:6 imposed); (iii) the max-entropy energy ratio equals the rank ratio only under H-degeneracy (equipartition application OPEN); (iv) the $\nu\to\tau$ reparametrization (A19.6 narrowed). v1.5 adds four ZHCS structural checks (§10): (v) the direct-sum projectors satisfy $P_c + P_b = I_{38}$ with ranks (32, 6); (vi) any block-connected 38-node graph has a one-dimensional Laplacian kernel (rank $L_\Gamma = 37$, ker = span$\{1_{38}\}$), verified on a representative connected graph — the corpus graph's connectivity (ZHCS-1) being the open finite computation; (vii) the harmonic charge gives $1^TP_c1 : 1^TP_b1 = 32 : 6$; (viii) a common rescaling $p_\star \to \alpha\, p_\star$ leaves Q_c/Q_b invariant, so the ratio evades A19.NG2. All 56 checks pass; the resulting honest statuses are: dust EOS IMPORTED-PROVEN, A19.NG2 PROVEN, BRL-5 OBSERVATIONALLY CONSISTENT, BRL-1/2/4 HYPOTHESIS-strong/OPEN, A19.5 IMPORTED-PROVEN precedent, the ZHCS rank-ratio PROVEN-CONDITIONAL on ZHCS-1 ∧ ZHCS-2, and the residual reduced to the four ZHCS theorems. v2.0 adds the ZHCS-1 polyhedral closure computation (four further checks, 60 total): the truncated-icosahedron face graph built from the icosahedron (12 pentagons of degree 5, 20 hexagons of degree 6) is connected (Laplacian rank 31); the cube face graph is connected (rank 5); the assembled 38-node seam graph has rank $L_\Gamma = 37$ (both with a complete-bipartite cross-block and with a single cross-edge); and the truncated-icosahedron vertex-face incidence (60×32) has a one-dimensional $B^T B$ kernel realizing ZS-F2 Lemma 4.5 ($v_p = -2v_h$ uniformly, nonzero on all 32 faces, $(\Sigma v)^2 = 4/17$ exactly), with the contrasting failure mode (mutually non-adjacent squares, single dipole cross-support) recorded at rank 33. With these, ZHCS-1 is DERIVED, ZHCS-2 is DERIVED-CONDITIONAL, and the 32:6 abundance ratio is DERIVED-CONDITIONAL on the single ZHCS-2 operator identification.

Appendix B. Dependencies and Cross-References

Table B1. Principal corpus dependencies.

Paper	Role in ZS-A19
ZS-F2	$\Omega_{cdm} = F(TI)/Q^2 = 32/121$; CDM as geometric boundary tension (the definition implemented here).
ZS-A18 (v1.5)	Massless-Goldstone NO-GO; frozen, not modified; defines the scope this carrier lies outside of.
ZS-Q11 / ZS-F23	Logical algebra $A_{ZS} = M_3 \oplus \mathbb{C} \oplus M_5$; canonical trace and II_1 embedding; Condition C, which by Theorem A19.NG1 is a distinct trace from the channel-trace Condition BT-C (not unified).
ZS-M9	Truncated-icosahedron 32-dimensional face-module decomposition; $A_5 \cong I$ structure.
ZS-S10	Vortex Bose/Fermi duality and the holonomy (not Dirac) nature of the core spinor (motivating the ontology shift).
ZS-A1 / ZS-S3	Massless Goldstone ε-Halo and Goldstone theorem, preserved ($m_\theta = 0$).
ZS-U3 / ZS-F5	Baryon asymmetry $\eta_B = (6/11)^{35} = 6.117\times10^{-10}$ and the $\eta_B\to\omega_b$ consistency (Independence Warning); used by Theorem A19.4.

Paper	Role in ZS-A19
ZS-F0	GHY boundary term, Robin condition, BV-BFV functor, and J as a boundary constraint operator (not a bulk symmetry); supports S_match (§8.3).
ZS-U6	Note that radiation-era Stefan–Boltzmann equipartition does not transfer directly to the matter era; motivates the §8.7 reinterpretation.
ZS-S1	Z-sector Z_2 split into one Z_2-even (β_0) and one Z_2-odd gauge mode; basis for the XQ−1 = 32 seam projector (§8.1).
ZS-F10	The i-tetration internal-time axis $\nu = (A/\pi)\ln(t/t\_P)$, t_phase, t_strobo; ruled out as the dust clock by Theorem A19.6 but retained as the corpus information-time.
ZS-F11	Closure of an OPEN observer/time coordinate by re-reading existing PROVEN structures (zero new dof) — the methodological precedent for the §9.5 clock identification.

References

[1] A. H. Chamseddine and V. Mukhanov, "Mimetic Dark Matter," J. High Energy Phys. 11, 135 (2013); arXiv:1308.5410.

[2] A. H. Chamseddine, V. Mukhanov, and A. Vikman, "Cosmology with Mimetic Matter," J. Cosmol. Astropart. Phys. 06, 017 (2014); arXiv:1403.3961.

[3] S. Dubovsky, L. Hui, A. Nicolis, and D. T. Son, "Effective field theory for hydrodynamics: thermodynamics, and the derivative expansion," Phys. Rev. D 85, 085029 (2012); arXiv:1107.0731.

[4] S. Mukohyama, "Dark matter as integration constant in Hořava-Lifshitz gravity," Phys. Rev. D 80, 064005 (2009); arXiv:0905.3563.

[5] J. D. Brown and K. V. Kuchař, "Dust as a standard of space and time in canonical quantum gravity," Phys. Rev. D 51, 5600 (1995); arXiv:gr-qc/9409001.

[6] A. S. Cattaneo, P. Mnev, and N. Reshetikhin, "Classical BV theories on manifolds with boundary," Commun. Math. Phys. 332, 535 (2014); arXiv:1201.0290.

[7] A. S. Cattaneo, N. Moshayedi, and K. Wernli, "Globalization for perturbative quantization of nonlinear split AKSZ sigma models on manifolds with boundary," Commun. Math. Phys. 372, 213 (2019); arXiv:1807.11782. [modified quantum master equation (mQME)]

[8] R. P. Malik, "BRST cohomology and Hodge decomposition theorem in Abelian gauge theory," Int. J. Mod. Phys. A 15, 1685 (2000); arXiv:hep-th/9808040.

[9] W. Donnelly and L. Freidel, "Local subsystems in gauge theory and gravity," J. High Energy Phys. 09 (2016) 102; arXiv:1601.04744.

[10] C.-P. Ma and E. Bertschinger, "Cosmological perturbation theory in the synchronous and conformal Newtonian gauges," Astrophys. J. 455, 7 (1995); arXiv:astro-ph/9506072.

[11] E. T. Jaynes, "Information theory and statistical mechanics," Phys. Rev. 106, 620 (1957).

[12] A. Connes and C. Rovelli, "Von Neumann algebra automorphisms and time-thermodynamics relation in generally covariant quantum theories," Class. Quantum Grav. 11, 2899 (1994); arXiv:gr-qc/9406019.

[13] M. Barriola and A. Vilenkin, "Gravitational field of a global monopole," Phys. Rev. Lett. 63, 341 (1989).

[14] A. Vilenkin and E. P. S. Shellard, Cosmic Strings and Other Topological Defects (Cambridge University Press, Cambridge, 1994).

[15] E. Witten, "Superconducting strings," Nucl. Phys. B 249, 557 (1985).

[16] R. Jackiw and P. Rossi, "Zero modes of the vortex-fermion system," Nucl. Phys. B 190, 681 (1981).

[17] R. H. Brandenberger, B. Carter, A.-C. Davis, and M. Trodden, "Cosmic vortons and particle physics constraints," Phys. Rev. D 54, 6059 (1996); arXiv:hep-ph/9605382.

[18] B. Carter and A.-C. Davis, "Chiral vortons and cosmological constraints on particle physics," Phys. Rev. D 61, 123501 (2000).

[19] F. J. Murray and J. von Neumann, "On rings of operators," Ann. Math. 37, 116 (1936).

[20] A. Connes, "Classification of injective factors," Ann. Math. 104, 73 (1976).

[21] Planck Collaboration (N. Aghanim et al.), "Planck 2018 results. VI. Cosmological parameters," Astron. Astrophys. 641, A6 (2020); arXiv:1807.06209.

Version History

v1.0 (June 2026): Initial public release. Consolidated from internal Z-Spin Collaboration deep-exploration research notes (free-exploration sessions, June 2026) following the freezing of ZS-A18 at v1.5. Established the geometric-dust ontology, the equivariant boundary-rank–trace theorem, and the two NO-GO characterizations of the 32-charge; registered Condition BT-C and the consistency gaps O-A19.2 through O-A19.5.

v1.1 (June 2026): Revision incorporating an external technical review (free-exploration session log, June 2026). Principal changes, no new physics introduced: (1) added Theorem A19.NG1 (PROVEN NO-GO) — the unique faithful unital embedding A_ZS = $M_3 \oplus \mathbb{C} \oplus M_5 \to M_{11}$ yields center weights (3, 3, 5)/11 ≠ ZS-F23 (3, 2, 6)/11; (2) RETRACTED the v1.0 claim that one theorem closes both the gravitational-entropy normalization and the dark-matter abundance (NC-A19.5), replacing it with the §7 repair paths and a finite gate set; (3) demoted Theorem A19.1 from "DERIVED (external)" to existence IMPORTED-PROVEN with Z-Spin realization HYPOTHESIS-strong/OPEN, and separated the three external constructions as alternative precedents; (4) corrected the pullback pressure sign to $p = F - bF_b$; (5) restricted "no particle number / no mass scale" to "no microscopic particle relic abundance or formation temperature," and the banner from "Zero Free Parameters" to "zero new fitted parameters; μ, state, action-level origin OPEN"; (6) elevated O-A19.5 (A_5 vs (2,3,6)) to a structural tension and set the cosmological status of Theorem A19.2 to DERIVED-CANDIDATE; (7) clarified that P_BT is a rank-32 candidate, not a canonical projection, and that the no-equipartition statement is algebraic only; (8) removed the "single remaining physical bridge" language throughout; (9) added the ε-Halo double-counting version-conflict (O-A19.6) with three resolutions; (10) softened the §9 CAMB statement to require BTD-1–5 plus adiabatic initial conditions; (11) scoped the verification banner to 29/29 algebraic and arithmetic checks with 0/5 action-level cosmological gates closed. (A, Q, dim Z) = (35/437, 11, 2) LOCKED; no new free parameter.

v1.2 (June 2026): Revision introducing a more corpus-native action-level route, no new fitted parameter. Principal additions: (1) new §8 Boundary-Rank-Locked Z-Clock Dust (BRL-ZCD) — an $XQ-1 = 32$ seam projector, a constrained Z-clock dust action (Theorem A19.3, exact $w = c_s^2 = 0$, $\rho \propto a^{-3}$), and a baryon-anchored boundary lock fixing $\rho_c/\rho_b = 32/6$ with no free μ; (2) Theorem A19.4 (h-free CDM density): $\omega_c = (32/6)\,\omega_b = 0.119112$, with the parameter-free ratio $\omega_c/\omega_b = 16/3$; (3) a registered version-conflict reinterpreting ZS-F2's radiation-era equipartition step (§8.7); (4) the BRL-1–6 gate set superseding the BTD set on the primary route; (5) new non-claims and falsification gates F-A19.10, F-A19.11; (6) a second anti-numerology audit AN-A19.2; (7) verification banner scoped to 35/35 checks with 0/6 action-level gates closed. The §5 rank–trace and Theorem A19.NG1 retained; ZS-A18 frozen at v1.5. (A, Q, dim Z) = (35/437, 11, 2) LOCKED; no new free parameter.

v1.3 (June 2026): Gate-testing revision adding four computational results. New §9 reporting tests of the §8 gates; a CAMB acoustic-peak run; the S_3 uniqueness of the seam mode; the $\delta_c = \delta_b$ implication; and the i-tetration clock norm. NOTE: several v1.3 conclusions are CORRECTED in v1.4 below — in particular the claims that BRL-5 was "observationally VERIFIED," that equipartition was on a "PROVEN footing," and that three routes converge on a single emergence node. The verification suite was extended to 48 checks. (A, Q, dim Z) = (35/437, 11, 2) LOCKED; no new free parameter.

v1.4 (June 2026): Honesty-restoring revision incorporating a second external technical review of v1.2 and v1.3; no new physics or fitted parameters. Principal changes: (1) added Theorem A19.NG2 (PROVEN NO-GO, §8.2a) — for a scalar clock and scalar multiplier the seam projector enters the bulk action only through $\mathrm{Tr}\, P_c = 32$, removable by $\lambda_c \to \lambda_c/32$, so the rank-32 lock has no bulk effect; registered the boundary-Noether-charge and operator-valued repairs (G1) as OPEN; (2) RETRACTED the v1.3 single-emergence-node claim — by Theorem A19.NG1 the channel trace (BT-C) and the ZS-F23 modular weights (Condition C) are distinct — and replaced it with the honest gate inventory G1–G8 (§9.6); (3) downgraded Theorem A19.5 to an IMPORTED-PROVEN precedent with the Z-Spin application OPEN (three sub-gates: mode-energy degeneracy, freeze-out preserving 32:6, baryon thermal history); (4) downgraded BRL-5 from VERIFIED to COMPUTED / OBSERVATIONALLY CONSISTENT (standard-CDM transfer assumed, TT only, peak-shape not likelihood); (5) downgraded the dust clock (BRL-1) and the seam construction (BRL-2) to HYPOTHESIS-strong/OPEN (worldline proper time ≠ global scalar clock; X⊗Q not derived from the direct-sum register; rank absorbed), and BRL-4 to OPEN (gauge-invariant S_cb; p_b ≠ 0 at confinement; clock–lock link); (6) narrowed Theorem A19.6 (ν-specific; proper time recoverable by reparametrization); (7) noted S_match imposes any ratio ("conditionally yields" not "predicts"), and that conserved-current matching (G4) and a rank-6 baryon projector (G3) are missing; (8) fixed residual editorial conflicts with NG1 ("one theorem closes both," "same canonical trace that ZS-F23 derives," "unified with BT-C," "single unclosed link") and restored this version history; (9) overall status of BRL-ZCD set to BOOTSTRAP-HYPOTHESIS; verification suite extended to 52 checks (four review-driven obstruction proofs). The geometric dust EOS remains IMPORTED-PROVEN, the integers 32, 6, η_B pre-exist, and ZS-A18 remains frozen. Adding A19.NG2 raises the honesty of the program. (A, Q, dim Z) = (35/437, 11, 2) LOCKED; no new free parameter.

v1.5 (June 2026): Closure-program revision; no new physics or fitted parameters, and all v1.4 honesty corrections retained. Principal changes: (1) added §10, the ZHCS (Z-Seam Harmonic mass-Charge Splitting) boundary-charge closure program — the rank is relocated out of the bulk (per A19.NG2) onto a conserved boundary canonical charge on a connected 38-node graph of the 32 TI cold faces and the 6 tO baryon faces; if the ZS-F0 boundary variation makes the charge graph-harmonic (d_Γ p = 0) and the graph is connected, then 32:6 is the rescaling-invariant ratio of two projections of one parent charge (never written into the action), with the bulk supplied by Brown–Kuchař dust; (2) reduced the gate set G1–G8 to four minimal theorems ZHCS-1 (connectivity, rank L_Γ = 37 — immediately computable), ZHCS-2 (ZS-F0 variation ⟹ d_Γ p = 0 — the real bottleneck), ZHCS-3 (canonical gluing to the dust momentum), ZHCS-4 (single adiabatic parent clock); the rank-ratio theorem is PROVEN conditional on ZHCS-1 ∧ ZHCS-2, the program HYPOTHESIS-strong; (3) supplied the rank-6 baryon projector by a canonical direct sum E_c ⊕ E_b (G3, DERIVED-CONDITIONAL) and a gauge-invariant S_cb = 0; (4) resolved the residual v1.4 editorial conflicts the second review flagged — Table 3 statuses updated to v1.4 and relabelled as superseded, the §8.2 title split into the IMPORTED-PROVEN EOS / A19.NG2 / OPEN realization layers, the §8.1 BRL-2 sentence set to OPEN, the §8.7 ratio split into a DERIVED mode-count vs an OPEN/imposed energy-density ratio, §12 "predicts" changed to conditional numerical output, NC-A19.7 changed from a single link to the G1–G8 gate set, the Acknowledgements count 48 → 56, the VERIFIED legend reduced to its general definition (no claim at that level), and the falsification-table title corrected to v1.5; (5) retitled the paper to foreground the rank-absorption no-go and the boundary-charge program; (6) verification suite extended to 56 checks (four ZHCS structural checks). ZS-A18 remains frozen at v1.5. (A, Q, dim Z) = (35/437, 11, 2) LOCKED; no new free parameter.

v2.0 (June 2026): Deep-exploration closure of the two gates v1.5 left open; no new physics or fitted parameters, ZS-A18 frozen, (A, Q, dim Z) = (35/437, 11, 2) LOCKED. Principal changes: (1) ZHCS-1 closed from OPEN to DERIVED by explicit computation on the actual polyhedra — the truncated-icosahedron face graph (12 pentagons of degree 5, 20 hexagons of degree 6) and the cube face graph are each connected, the 38-node seam graph has Laplacian rank L_Γ = 37 (robust to using a single cross-edge), and the cross-block is shown to have full support by the PROVEN Euler null mode of the truncated-icosahedron vertex-face incidence (ZS-F2 Lemma 4.5: v_p = −2v_h, $(\Sigma v)^2$ = 4/17, nonzero on all 32 faces); the contrasting failure mode (non-adjacent squares, single-dipole cross-support → rank 33) is recorded for honesty; (2) ZHCS-2 closed from OPEN to DERIVED-CONDITIONAL — the ZS-F0 BV-BFV modified quantum master equation (Ω_∂Σψ = 0, IMPORTED Cattaneo–Moshayedi–Wernli), the BRST Hodge decomposition (harmonic = closed ∩ co-closed, IMPORTED Malik), and the seam J as a boundary constraint operator (PROVEN) force the boundary charge to be a harmonic 0-cochain, leaving only the identification Ω_∂Σ ≅ d_Γ on the 38-face basis; (3) consequently the rank-ratio theorem and the 32:6 abundance ratio rise from PROVEN-CONDITIONAL on two open gates to DERIVED-CONDITIONAL on the single ZHCS-2 identification, with A19.NG2 respected throughout; (4) ZHCS-3/4 remain OPEN as the registered residuals for the full perturbation spectrum, so v2.0 reports a conditional derivation and explicitly not an unconditional one; (5) verification suite extended to 60 checks (the ZHCS-1 polyhedral closure computation), with the companion script zhcs_closure_test.py; (6) retitled to foreground the lift of the 32/121 ratio to DERIVED-CONDITIONAL; references for the mQME and BRST-Hodge results added.